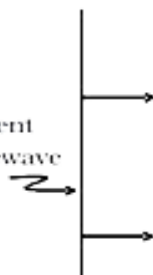
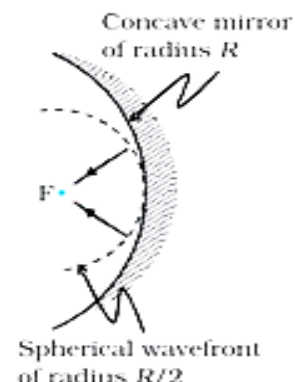
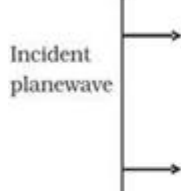
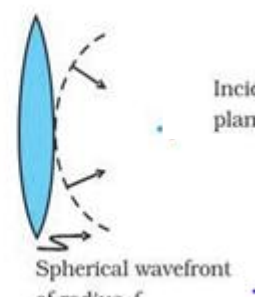


| <b>Marking Scheme</b><br><b>Strictly Confidential</b><br><b>(For Internal and Restricted use only)</b><br><b>Senior School Certificate Examination, 2023</b><br><b>SUBJECT NAME PHYSICS (PAPER CODE 55/3/1)</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><u>General Instructions: -</u></b>                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>1</b>                                                                                                                                                                                                        | You are aware that evaluation is the most important process in the actual and correct assessment of the candidates. A small mistake in evaluation may lead to serious problems which may affect the future of the candidates, education system and teaching profession. To avoid mistakes, it is requested that before starting evaluation, you must read and understand the spot evaluation guidelines carefully.                                                                                                                                                                                                                                                                         |
| <b>2</b>                                                                                                                                                                                                        | <b>“Evaluation policy is a confidential policy as it is related to the confidentiality of the examinations conducted, Evaluation done and several other aspects. Its’ leakage to public in any manner could lead to derailment of the examination system and affect the life and future of millions of candidates. Sharing this policy/document to anyone, publishing in any magazine and printing in News Paper/Website etc may invite action under various rules of the Board and IPC.”</b>                                                                                                                                                                                              |
| <b>3</b>                                                                                                                                                                                                        | Evaluation is to be done as per instructions provided in the Marking Scheme. It should not be done according to one’s own interpretation or any other consideration. Marking Scheme should be strictly adhered to and religiously followed. <b>However, while evaluating, answers which are based on latest information or knowledge and/or are innovative, they may be assessed for their correctness otherwise and due marks be awarded to them. In class-X, while evaluating two competency-based questions, please try to understand given answer and even if reply is not from marking scheme but correct competency is enumerated by the candidate, due marks should be awarded.</b> |
| <b>4</b>                                                                                                                                                                                                        | The Marking scheme carries only suggested value points for the answers. These are in the nature of Guidelines only and do not constitute the complete answer. The students can have their own expression and if the expression is correct, the due marks should be awarded accordingly.                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>5</b>                                                                                                                                                                                                        | The Head-Examiner must go through the first five answer books evaluated by each evaluator on the first day, to ensure that evaluation has been carried out as per the instructions given in the Marking Scheme. If there is any variation, the same should be zero after deliberation and discussion. The remaining answer books meant for evaluation shall be given only after ensuring that there is no significant variation in the marking of individual evaluators.                                                                                                                                                                                                                   |
| <b>6</b>                                                                                                                                                                                                        | Evaluators will mark( $\sqrt{\phantom{x}}$ ) wherever answer is correct. For wrong answer CROSS ‘X’ be marked. Evaluators will not put right (✓) while evaluating which gives an impression that answer is correct and no marks are awarded. <b>This is most common mistake which evaluators are committing.</b>                                                                                                                                                                                                                                                                                                                                                                           |
| <b>7</b>                                                                                                                                                                                                        | If a question has parts, please award marks on the right-hand side for each part. Marks awarded for different parts of the question should then be totaled up and written in the left-hand margin and encircled. This may be followed strictly.                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>8</b>                                                                                                                                                                                                        | If a question does not have any parts, marks must be awarded in the left-hand margin and encircled. This may also be followed strictly.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>9</b>                                                                                                                                                                                                        | If a student has attempted an extra question, answer of the question deserving more marks should be retained and the other answer scored out with a note <b>“Extra Question”</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>10</b>                                                                                                                                                                                                       | No marks to be deducted for the cumulative effect of an error. It should be penalized only once.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>11</b>                                                                                                                                                                                                       | A full scale of marks 0-70(example 0 to 80/70/60/50/40/30 marks as given in Question Paper) has to be used. Please do not hesitate to award full marks if the answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           | deserves it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>12</b> | Every examiner has to necessarily do evaluation work for full working hours i.e., 8 hours every day and evaluate 20 answer books per day in main subjects and 25 answer books per day in other subjects (Details are given in Spot Guidelines). This is in view of the reduced syllabus and number of questions in question paper.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>13</b> | <p>Ensure that you do not make the following common types of errors committed by the Examiner in the past:-</p> <ul style="list-style-type: none"> <li>• Leaving answer or part thereof unassessed in an answer book.</li> <li>• Giving more marks for an answer than assigned to it.</li> <li>• Wrong totaling of marks awarded on an answer.</li> <li>• Wrong transfer of marks from the inside pages of the answer book to the title page.</li> <li>• Wrong question wise totaling on the title page.</li> <li>• Wrong totaling of marks of the two columns on the title page.</li> <li>• Wrong grand total.</li> <li>• Marks in words and figures not tallying/not same.</li> <li>• Wrong transfer of marks from the answer book to online award list.</li> <li>• Answers marked as correct, but marks not awarded. (Ensure that the right tick mark is correctly and clearly indicated. It should merely be a line. Same is with the X for incorrect answer.)</li> <li>• Half or a part of answer marked correct and the rest as wrong, but no marks awarded.</li> </ul> |
| <b>14</b> | While evaluating the answer books if the answer is found to be totally incorrect, it should be marked as cross (X) and awarded zero (0) Marks.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>15</b> | Any un assessed portion, non-carrying over of marks to the title page, or totaling error detected by the candidate shall damage the prestige of all the personnel engaged in the evaluation work as also of the Board. Hence, in order to uphold the prestige of all concerned, it is again reiterated that the instructions be followed meticulously and judiciously.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>16</b> | The Examiners should acquaint themselves with the guidelines given in the “ <b>Guidelines for spot Evaluation</b> ” before starting the actual evaluation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>17</b> | Every Examiner shall also ensure that all the answers are evaluated, marks carried over to the title page, correctly totaled and written in figures and words.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>18</b> | The candidates are entitled to obtain photocopy of the Answer Book on request on payment of the prescribed processing fee. All Examiners/Additional Head Examiners/Head Examiners are once again reminded that they must ensure that evaluation is carried out strictly as per value points for each answer as given in the Marking Scheme.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

| MARKING SCHEME: PHYSICS(042) |                                                                                                             |       |             |
|------------------------------|-------------------------------------------------------------------------------------------------------------|-------|-------------|
| Code: 55/3/1                 |                                                                                                             |       |             |
| Q.No.                        | VALUE POINTS/EXPECTED ANSWERS                                                                               | Marks | Total Marks |
|                              | SECTION A                                                                                                   |       |             |
| 1.                           | (b) $-(1.0 \times 10^3 \frac{N}{C})\hat{i}$                                                                 | 1     | 1           |
| 2.                           | (a) Electric Field                                                                                          | 1     | 1           |
| 3.                           | (d) $nev_d$                                                                                                 | 1     | 1           |
| 4.                           | (c) <div data-bbox="321 632 540 846"> </div>                                                                | 1     | 1           |
| 5.                           | (d) a helix with increasing pitch.                                                                          | 1     | 1           |
| 6.                           | (b) inductor decreases and the capacitor increases                                                          | 1     | 1           |
| 7.                           | (d) ultraviolet                                                                                             | 1     | 1           |
| 8.                           | (c) $I \propto A^2$                                                                                         | 1     | 1           |
| 9.                           | (b) twice                                                                                                   | 1     | 1           |
| 10.                          | (a) h                                                                                                       | 1     | 1           |
| 11.                          | (b) 0.51eV                                                                                                  | 1     | 1           |
| 12.                          | (b) $\frac{9}{5}$                                                                                           | 1     | 1           |
| 13.                          | (c) $5 \times 10^9 \text{ m}^{-3}$                                                                          | 1     | 1           |
| 14.                          | (c) the potential barrier is raised                                                                         | 1     | 1           |
| 15.                          | (a) $\frac{V_o}{\sqrt{2}}$                                                                                  | 1     | 1           |
| 16.                          | (d) Assertion (A) is false and Reason (R) is also false.                                                    | 1     | 1           |
| 17.                          | (a) Both Assertion(A) and Reason (R) are true and Reason(R) is the correct explanation of the Assertion (A) | 1     | 1           |
| 18.                          | (c) Assertion (A) is true, but Reason (R) is false                                                          | 1     | 1           |

|     | SECTION B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                             |   |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|---|
| 19. | <div>Ratio of magnetic fields at the centres of loop A &amp; loop B 2</div> $B = \frac{\mu_o NI}{2r}$ $l = 2\pi r ; \quad 2\pi r_A = 3(2\pi r_B)$ $r_B = \frac{r_A}{3}$ $\frac{B_A}{B_B} = \frac{\mu_o N_A I}{2r_A} \times \frac{2r_B}{\mu_o N_B I}$ $\frac{B_A}{B_B} = \frac{1}{9}$                                                                                                                                                                                                                                                                                                                                                       | <p>1/2</p> <p>1/2</p> <p>1/2</p> <p>1/2</p> | 2 |
| 20. | <div>Definition of Displacement Current 1</div> <div>Difference 1</div> <p>Displacement current: It is the current that arises due to the rate of change of electric field/flux.</p> <p><b>Alternatively:-</b></p> $I_d = \epsilon_0 \left( \frac{d\phi_E}{dt} \right)$ <p><b>Alternatively:</b> The term with units of current to explain the continuity of current in a region.</p> <p><b>Difference:</b></p> <p>Displacement current is due to change in electric flux.<br/>Conduction current is due to flow of electrons.</p> <p><b>Alternatively:</b></p> $I_d = \epsilon_0 \left( \frac{d\phi_E}{dt} \right)$ $I_c = \frac{dq}{dt}$ | <p>1</p> <p>1</p>                           | 2 |
| 21. | <p>(a)</p> <div>Statement of Huygen's Principle 1</div> <div>Explanation 1</div> <p>Statement: Each point of the wavefront is the source of secondary disturbance in all directions.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>1/2</p>                                  |   |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                            |                            |   |      |                         |   |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------------------|---|------|-------------------------|---|--|
| <p>Common tangent to all the secondary wavelets gives new position of the wavefront.</p> <p>Explanation: Light energy cannot travel in backward direction.</p> <p><b>Alternatively:</b></p> <p>It was an adhoc assumption .</p> <p><b>Alternatively:</b></p> <p>For back wave: <math>I = \frac{1}{2} (1+ \cos \theta)</math></p> <p>at <math>\theta =180^{\circ}</math> ; contribution is zero.</p> <p><b>Alternatively:</b></p> <p>Amplitude of secondary wavelets is maximum in forward direction and zero in backward in direction.</p> <p><b>Note:</b> If any other relevant explanation given, give full credit.</p> | <p>½</p> <p>1</p>          |                            |   |      |                         |   |  |
| <p style="text-align: center;"><b>OR</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                            |                            |   |      |                         |   |  |
| <p>(b)</p> <table><tr><td>(i)</td><td>Diagram for concave mirror</td><td>1</td></tr><tr><td>(ii)</td><td>Diagram for convex lens</td><td>1</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                              | (i)                        | Diagram for concave mirror | 1 | (ii) | Diagram for convex lens | 1 |  |
| (i)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Diagram for concave mirror | 1                          |   |      |                         |   |  |
| (ii)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Diagram for convex lens    | 1                          |   |      |                         |   |  |
| <p>(i)</p> <div><div><p>Incident plane wave</p></div><div><p>Concave mirror of radius R</p></div></div>                                                                                                                                                                                                                                                                                                                                             | <p>1</p>                   |                            |   |      |                         |   |  |
| <p>(ii)</p> <div><div><p>Incident plane wave</p></div><div></div></div>                                                                                                                                                                                                                                                                                                                                                                            | <p>1</p>                   |                            |   |      |                         |   |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>2</p>                   |                            |   |      |                         |   |  |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 22. | <div data-bbox="297 237 1307 300" style="border: 1px solid black; padding: 5px; margin-bottom: 10px;"> Calculation of critical angle <span style="float: right;">2</span> </div> <p><b>From Snell 's law:-</b><br/> <math>\mu_A \sin i_c = \mu_B \sin 90^\circ</math><br/> <math>2 \times \sin i_c = \sqrt{2} \times 1</math><br/> <math>\sin i_c = \frac{1}{\sqrt{2}}</math><br/> <math>i_c = 45^\circ</math></p> <p><b>Alternatively:</b><br/> <math>\sin i_c = \frac{1}{\mu_B}</math><br/> <math>\sin i_c = \frac{1}{\sqrt{2}}</math><br/> <math>i_c = 45^\circ</math></p>                                                                                                                                                                                                                                                                                                                       | <div data-bbox="1352 373 1385 405"><math>\frac{1}{2}</math></div> <div data-bbox="1352 447 1385 478"><math>\frac{1}{2}</math></div> <div data-bbox="1352 520 1385 552"><math>\frac{1}{2}</math></div> <div data-bbox="1352 594 1385 625"><math>\frac{1}{2}</math></div> <div data-bbox="1352 783 1385 814">1</div> <div data-bbox="1352 856 1385 888"><math>\frac{1}{2}</math></div> <div data-bbox="1352 930 1385 961"><math>\frac{1}{2}</math></div> | 2 |
| 23. | <div data-bbox="297 1071 1307 1150" style="border: 1px solid black; padding: 5px; margin-bottom: 10px;"> Graph of binding energy per nucleon as a function of mass number A <span style="float: right;">1</span><br/> Explanation <span style="float: right;">1</span> </div> <div data-bbox="332 1197 1278 1606"> <p style="text-align: center;">Mass number (A)</p> </div> <p><b>Explanation:-</b> Nuclear forces are short range &amp; show saturation, while the electrostatic force are neither short range nor show any saturation. Hence for heavier nuclei (<math>A &gt; 170</math>) the electrostatic force of repulsion becomes predominant, decreasing the binding energy per nucleon.</p> <p><b>Alternatively:-</b> As the size of the nucleus increases, the nucleus becomes unstable.</p> <p><b>Note:</b> No deduction of marks if values of elements are not shown in the graph.</p> | <div data-bbox="1352 1371 1385 1402">1</div> <div data-bbox="1352 1665 1385 1696">1</div>                                                                                                                                                                                                                                                                                                                                                              |   |

|                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                       |     |                                                                                                                    |   |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|-----|--------------------------------------------------------------------------------------------------------------------|---|
|                                                                       | <p style="text-align: center;"><b>OR</b></p> <p>(b)</p> <table border="1"> <tr> <td>Expression for radius of the n<sup>th</sup> orbit in a hydrogen atom</td> <td style="text-align: right;">2</td> </tr> </table> $\frac{mv^2}{r_n} = \frac{kq^2}{r_n^2} = \frac{e^2}{4\pi \epsilon_o r_n^2} \quad \text{-----(1)}$ $mvr_n = \frac{nh}{2\pi} \quad \text{-----(2)}$ <p>Using equation (1) &amp;(2)</p> $r_n = \frac{n^2 h^2 4\pi \epsilon_o}{m(2\pi)^2 e^2} = 0.53 \times 10^{-10} n^2 \text{ m}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Expression for radius of the n <sup>th</sup> orbit in a hydrogen atom | 2   | <p style="text-align: center;">1/2</p> <p style="text-align: center;">1/2</p> <p style="text-align: center;">1</p> | 2 |
| Expression for radius of the n <sup>th</sup> orbit in a hydrogen atom | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                       |     |                                                                                                                    |   |
| 24.                                                                   | <table border="1"> <tr> <td>Roles of diffusion current and drift current</td> <td style="text-align: right;">1+1</td> </tr> </table> <p>During the formation of p-n junction ,and due to the concentration gradient across p-,and n-sides, holes diffuse from p-side to n-side (p → n) and electrons diffuse from n-side to p-side(n → p).When an electron diffuses from (n → p), it leaves behind an ionized donor(positive charge) on n-side which is immobile. Similarly, when a hole diffuses from (p → n) due to the concentration gradient, it leaves behind an ionised acceptor (negative charge) which is immobile. This space–charge region on either side of the junction together is known as depletion region.</p> <p>As a result, an electric field is developed across the junction. Due to this field, an electron on p-side of the junction moves to n-side and a hole on n-side of the junction moves to p-side. The motion of charge carriers due to the electric field is called drift .Initially, diffusion current is large and drift current is small.</p> <p>As the diffusion process continues, the electric field strength &amp; hence drift current increases .This process continues till diffusion &amp; drift current becomes equal.</p> | Roles of diffusion current and drift current                          | 1+1 | <p style="text-align: center;">1</p> <p style="text-align: center;">1</p>                                          | 2 |
| Roles of diffusion current and drift current                          | 1+1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                       |     |                                                                                                                    |   |



|                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                          |   |                                  |   |                                                                                                          |  |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|---|----------------------------------|---|----------------------------------------------------------------------------------------------------------|--|
|                                          | <p>(ii) <math>v_d = \left( \frac{eE}{m} \right) \tau = \frac{e\tau}{m} \left( \frac{V}{l} \right)</math></p> <p><math>v_d \propto V</math> ; <math>v'_d = 2 v_d</math></p> <p>(iii) Current density <math>J = \frac{I}{A} = \frac{V}{RA}</math></p> <p><math>\therefore J</math> is also doubled.</p> <p>(a) When <math>l</math> is halved</p> <p>(i) <math>E = \frac{V}{l}</math> ; <math>\therefore E' = 2E</math></p> <p>(ii) <math>v_d = \left( \frac{eE}{m} \right) \tau = \frac{e}{m} \left( \frac{V}{l} \right) \tau</math></p> <p><math>\therefore v'_d = 2 v_d</math></p> <p>(iii) <math>J = \frac{I}{A} = \frac{V}{RA} = \frac{V}{A} \left( \frac{A}{\rho l} \right) = \frac{V}{\rho l}</math></p> <p><math>\therefore J</math> is also doubled.</p>                                                                                                                                                                                                                                                                                                                                                                                        |                                          | 3 |                                  |   |                                                                                                          |  |
| 27.                                      | <table border="1"> <tr> <td>Definition /Meaning of Mutual Inductance</td> <td>1</td> </tr> <tr> <td>Expression for Mutual Inductance</td> <td>2</td> </tr> </table> <p><u>Mutual Inductance</u>-It is the magnetic flux in the secondary coil due to the flow of unit current in the primary coil.</p> <p><b>Alternatively:</b><br/>It is the emf induced in the secondary coil when rate of change of current in the primary coil is unity.</p> <p><u>Expression for Mutual Inductance</u><br/>Let <math>N_1</math> and <math>N_2</math> be the total number of turns of coils <math>S_1</math> &amp; <math>S_2</math>, respectively.</p> <p>When a current <math>I_2</math> is set up through <math>S_2</math>, it in turn sets up a magnetic flux through <math>S_1 \Rightarrow N_1 \phi_1 = M_{12} I_2</math> -----(1)</p> <p><math>M_{12}</math> is the mutual inductance of solenoid <math>S_1</math> w.r.t. <math>S_2</math>.</p> <p>Magnetic field due to <math>I_2</math> in <math>S_2 = \mu_o \left( \frac{N_2}{l} \right) I_2</math></p> <p><math>N_1 \phi_1 = (N_1)(\pi r_1^2) \left( \mu_o \frac{N_2}{l} I_2 \right)</math> -----(2)</p> | Definition /Meaning of Mutual Inductance | 1 | Expression for Mutual Inductance | 2 | <p>1</p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> |  |
| Definition /Meaning of Mutual Inductance | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                          |   |                                  |   |                                                                                                          |  |
| Expression for Mutual Inductance         | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                          |   |                                  |   |                                                                                                          |  |

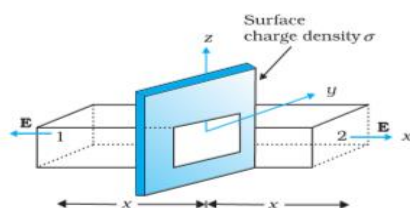


|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                 |   |                      |   |                                                                                                                                                                 |  |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----------------------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|                           | <p>Instantaneous Power;</p> $P = VI = (V_m \sin \omega t) \times I_m \sin(\omega t + \phi)$ $P = \frac{V_m I_m}{2} [\cos \phi - \cos(2\omega t + \phi)] \quad \text{----(1)}$ <p>The average power over a cycle is given by the average of the two terms in the R.H.S of equation (1). It is only the second term which is time dependent .Its average is zero (the positive half of the cosine cancels the negative half). Therefore,</p> $P_{avg} = \frac{V_m I_m}{2} \cos \phi = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos \phi$ $P_{avg} = V_{rms} I_{rms} \cos \phi$ <p><b>Alternatively:-</b><br/>If the expression is deduced using integration, then full credit to be given.</p> <p>(i) Power factor for purely inductive circuit, <math>\phi = \frac{\pi}{2} \Rightarrow \cos \phi = 0</math></p> <p>(ii) Power factor for purely resistive circuit; <math>\phi = 0 \Rightarrow \cos \phi = 1</math></p> | <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> | 3 |                      |   |                                                                                                                                                                 |  |
| 29.                       | <table border="1"> <tr> <td>Calculation of wavelength</td> <td>2</td> </tr> <tr> <td>Effect on wavelength</td> <td>1</td> </tr> </table> $\lambda = \frac{h}{\sqrt{2mK}}$ $K = \left( \frac{500}{1.673} \right) eV$ $= \frac{500 \times 1.6 \times 10^{-19}}{1.673} J$ $\lambda_p = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 1.673 \times 10^{-27} \times \left( \frac{500 \times 1.6 \times 10^{-19}}{1.673} \right)}}$ $= 1.65 \times 10^{-12} m$ $\lambda \propto \frac{1}{\sqrt{m}}$                                                                                                                                                                                                                                                                                                                                                                                                                              | Calculation of wavelength                                                                                                                                                                       | 2 | Effect on wavelength | 1 | <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> |  |
| Calculation of wavelength | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                 |   |                      |   |                                                                                                                                                                 |  |
| Effect on wavelength      | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                 |   |                      |   |                                                                                                                                                                 |  |



|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |   |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---|
| <b>OR</b>        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |   |
| (b)              | <div style="border: 1px solid black; padding: 5px;">           (i) Graph to show the variation of the number of scattered particles as a function of scattering angle. <span style="float: right;">1</span><br/>           (ii) Two conclusions <span style="float: right;"><math>\frac{1}{2} + \frac{1}{2}</math></span><br/>                   Discovery of nucleus <span style="float: right;">1</span> </div>                                                                                                                                                                                                                                                                                                                                  |               |   |
| (i)              | <div style="text-align: center;"> </div> <p>(ii)- The entire positive charge and most of the mass of the atom are concentrated in a small space.</p> <p>-Many of the <math>\alpha</math>-particles pass through the foil. It means that they do not suffer any collisions.</p> <p>To deflect the <math>\alpha</math>-particle backwards, a large repulsive force is required, which is provided only if the greater part of the mass of the atom &amp; its positive charge were concentrated tightly at its centre. This led to the discovery of the nucleus in the atom.</p>                                                                                                                                                                      | 1             |   |
|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\frac{1}{2}$ |   |
|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\frac{1}{2}$ |   |
|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1             | 3 |
| <b>SECTION D</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |   |
| 31.              | <div style="border: 1px solid black; padding: 5px;">           (a)<br/>           (i) Definition &amp; SI Unit of Electric Flux <span style="float: right;"><math>\frac{1}{2} + \frac{1}{2}</math></span><br/>           (ii) Deriving the expression for electric field due to a uniformly charged infinite plane sheet. <span style="float: right;">2</span><br/>           (iii) Net charge enclosed by the cube <span style="float: right;">2</span> </div> <p>(i) <math>\phi = \vec{E} \cdot \vec{A}</math></p> <p><b>Alternatively:</b> Electric flux is the number of electric field lines passing through an area normally.</p> <p>S.I. unit of electric flux <math>\text{Nm}^2/\text{C}</math> or <math>\text{V}\cdot\text{m}</math>.</p> | $\frac{1}{2}$ |   |
|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\frac{1}{2}$ |   |

(ii)



1/2

From Gauss's law:-  $\phi = \oint \vec{E} \cdot d\vec{A} = \frac{q}{\epsilon_0}$

1/2

$$2EA = \frac{\sigma A}{\epsilon_0}$$

1/2

$$E = \frac{\sigma}{2\epsilon_0}$$

1/2

**Alternatively:** If the shape of the Gaussian surface is taken cylindrical, full credit to be given.

(iii)

$$\phi_L = E ds \cos 180^\circ = -E ds$$

1/2

$$= -BL^2$$

$$\phi_R = E ds \cos 0^\circ = E ds$$

1/2

$$= (AL + B)L^2 = AL^3 + BL^2$$

$$\text{Net flux} = \phi_L + \phi_R$$

$$= (AL^3 + BL^2) - BL^2$$

1/2

$$\text{Net flux} = AL^3 = \frac{q}{\epsilon_0}$$

$$\Rightarrow q = AL^3 \epsilon_0$$

1/2

**OR**

(b)

|                                                         |           |
|---------------------------------------------------------|-----------|
| (i) Definitions & S.I. Unit of electric potential       | 1/2 + 1/2 |
| (ii) Derivation of expression of Equivalent capacitance | 2         |
| (iii) Calculation of Electrostatic Potential Energy     | 2         |

(i) Electrical Potential – Electrostatic potential at any point in a region with electrostatic field is the work done in bringing a unit positive charge (without acceleration) from infinity to that point.

1/2

**Alternatively:-**

$$V = \frac{\text{Work Done}}{q}$$

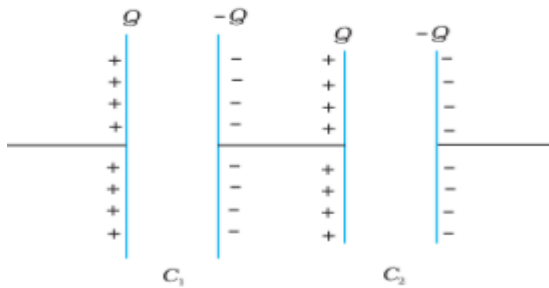
$$V = -\int \vec{E} \cdot d\vec{l}$$

S.I. unit of electrostatic potential is volt.

1/2

**Alternatively:-**

S.I. unit is J/C.



1/2

$$V = V_1 + V_2 = \frac{Q}{C_1} + \frac{Q}{C_2}$$

1/2

$$\frac{Q}{C_{eq.}} = Q \left( \frac{1}{C_1} + \frac{1}{C_2} \right)$$

1/2

$$\frac{1}{C_{eq.}} = \frac{1}{C_1} + \frac{1}{C_2}$$

1/2

(iii)

$$\text{Potential energy of the system} = K \left[ \frac{Q(-q)}{4a} + \frac{Qq}{3a} - \frac{q^2}{5a} \right]$$

1/2

Potential energy of the system = 0

$$\Rightarrow K \left[ \frac{-Qq}{4a} + \frac{Qq}{3a} - \frac{q^2}{5a} \right] = 0$$

1/2

$$\Rightarrow \frac{-Q}{4} + \frac{Q}{3} - \frac{q}{5} = 0$$

$$\Rightarrow +\frac{Q}{12} - \frac{q}{5} = 0$$

$$\Rightarrow Q = + \frac{12q}{5}$$

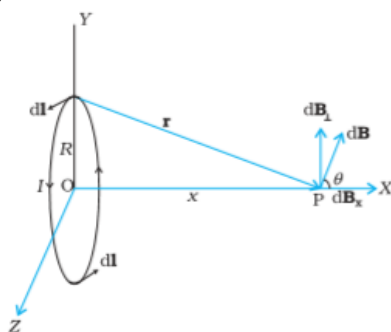
1

5

32.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                    |  |           |   |         |   |                                  |   |                              |   |                              |   |                                                                                                                                                                                                                                                    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| <p>(a)</p> <table border="1"><tr><td>(i) For a moving coil galvanometer</td><td></td></tr><tr><td>Principle</td><td>1</td></tr><tr><td>Working</td><td>1</td></tr><tr><td>Reason it cannot be used as such</td><td>1</td></tr><tr><td>(ii) Reason for radial field</td><td>1</td></tr><tr><td>How radial field is achieved</td><td>1</td></tr></table> <p>(i) Principle – When a rectangular loop carrying current <math>I</math> is placed in a uniform magnetic field, it experiences a torque.</p> <p>Working:-<br/>When a current flows through the coil of a galvanometer, a torque acts on it.</p> <p><math>\tau = NiAB \sin \theta</math><br/>For radial magnetic field; <math>\sin \theta = 1</math><br/>The spring provides a counter or restoring toqrue <math>k\phi</math>.</p> <p><math>k\phi = NiAB</math></p> <p><b>In equilibrium;</b> <math>\phi = \left( \frac{NAB}{k} \right) i</math></p> <p>Galvanometer cannot be used as such to measure current because:<br/>-It has large resistance and hence will change the value of current in the circuit.<br/>-It is a sensitive device.<br/>(Any one of the above)</p> <p>(ii) The magnetic field is made radial in a moving coil galvanometer so that the magnetic dipole moment (<math>\vec{m}</math>) is always perpendicular to the magnetic field (<math>\vec{B}</math>) Hence, <math>\sin \theta = 1</math> always<br/><b>Alternatively:</b> The magnetic field is made radial in a moving coil galvanometer to make the scale linear.</p> <p>It is achieved by using curved magnetic poles.</p> <p><b>Alternatively:-</b>By using soft iron cylindrical core.</p> | (i) For a moving coil galvanometer |  | Principle | 1 | Working | 1 | Reason it cannot be used as such | 1 | (ii) Reason for radial field | 1 | How radial field is achieved | 1 | 1<br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br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| (i) For a moving coil galvanometer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     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| Principle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| Reason it cannot be used as such                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| (ii) Reason for radial field                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           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| How radial field is achieved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           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(i)



$$dB = \frac{\mu_0}{4\pi} \frac{I |d\mathbf{l} \times \mathbf{r}|}{r^3}$$

$$= \frac{\mu_0 i dl \sin 90^\circ}{4\pi (x^2 + R^2)^{3/2}}$$

$dB_{\perp}$  cancels out.

$$\text{Net } B = \int dB_{\parallel} = \int dB \cos \theta$$

$$= \frac{\mu_0}{4\pi} \int \frac{idl}{(x^2 + R^2)^{3/2}} \times \frac{R}{(x^2 + R^2)^{1/2}}$$

$$= \frac{\mu_0 i R}{4\pi (x^2 + R^2)^{3/2}} \int dl$$

$$= \frac{\mu_0 i R}{4\pi (x^2 + R^2)^{3/2}} (2\pi R)$$

$$\mathbf{B} = B_{\parallel} \hat{\mathbf{i}} = \frac{\mu_0 I R^2}{2 (x^2 + R^2)^{3/2}} \hat{\mathbf{i}}$$

(ii) Differences

Diamagnetic Materials

(i) Susceptibility is between -1 and 0.

(ii) Relative permeability is between 0 and 1.

(iii)  $\mu < \mu_0$ 

(iv) Tendency to move from stronger to weaker part of external magnetism.

(v) is repelled by a magnet.

(vi) Field inside the material is reduced.

Paramagnetic Materials

(i) Susceptibility is a small positive number. (slightly greater than zero.)

(ii) Relative permeability is slightly greater than 1.

(iii)  $\mu > \mu_0$ 

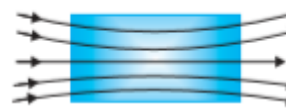
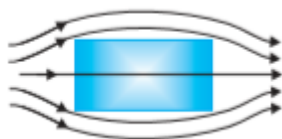
(iv) Tendency to move from region of weak to strong magnetic field.

(v) is weakly attracted by a magnet.

(vi) Field inside is slightly enhanced.

 $\frac{1}{2}$  $\frac{1}{2}$  $\frac{1}{2}$  $\frac{1}{2}$  $\frac{1}{2}$  $\frac{1}{2}$ 

1+1



Any two of the above mentioned differences.

5

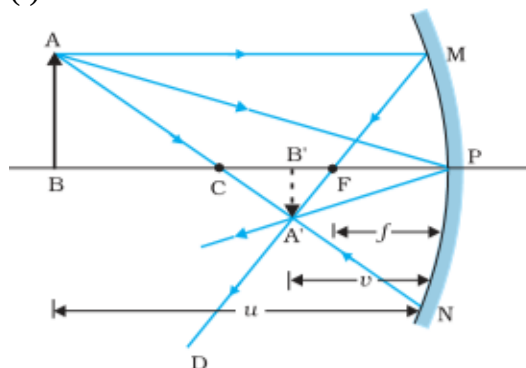
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**33.**

(a)

- |                                                                      |   |
|----------------------------------------------------------------------|---|
| (i) Ray diagram showing formation of real image in a concave mirror. | 1 |
| Obtaining the relation between $u, v$ and $R$                        | 2 |
| (ii) Position of image formed                                        | 1 |
| Height of image formed                                               | 1 |

(i)



From Fig. the two right-angled triangles  $A'B'F$  and  $MPF$  are similar. (For paraxial rays,  $MP$  can be considered to be a straight line perpendicular to  $CP$ .) Therefore,

$$\frac{B'A'}{PM} = \frac{B'F}{FP}$$

$$\text{or } \frac{B'A'}{BA} = \frac{B'F}{FP} \quad (\because PM = AB)$$

----- (i)

Since  $\angle APB = \angle A'PB'$ , the right angled triangles  $A'B'P$  and  $ABP$  are also similar. Therefore,

$$\frac{B'A'}{BA} = \frac{B'P}{BP}$$

----- (ii)

Comparing equations (i) and (ii)

$$\frac{B'F}{FP} = \frac{B'P - FP}{FP} = \frac{B'P}{BP}$$

-----(iii)

$B'P = -v$ ,  $FP = -f$ ,  $BP = -u$ ;

Using these in Eq.(iii) we get  $\frac{1}{v} + \frac{1}{u} = \frac{1}{f} = \frac{2}{R}$

**Alternatively:-** If the result derived by any other method, full credit to be given.

(ii) For lens:  $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$$u = -5m; \quad f = +1m$$

$$\frac{1}{v} - \frac{1}{-5} = \frac{1}{(+1)}$$

$$\Rightarrow v = \frac{5}{4}m = 1.25m$$

$$m = \frac{I}{O} = \frac{v}{u} = \frac{\left(+\frac{5}{4}\right)}{(-5)}$$

$$I = (-0.25) \times (1.8)$$

$$I = -0.45 \text{ m}$$

**OR**

(b)

(i) Ray diagram showing refraction of a ray of light through a rectangular glass prism.

1

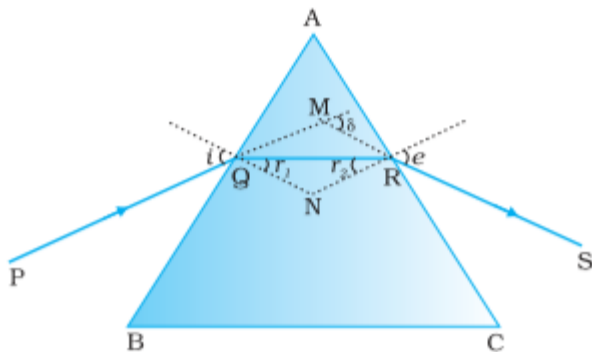
Obtaining the relation between  $\mu$ ,  $A$  &  $\delta_m$

2

(ii) Finding Refractive Index of material of the lens.

2

(i)

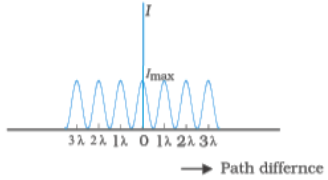
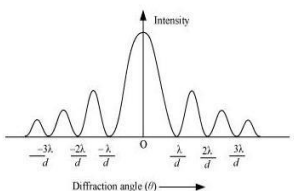


1

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                 |   |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|     | <p>In the quadrilateral AQNR, two of the angles (at the vertices Q and R) are right angles. Therefore, the sum of the other angles of the quadrilateral is <math>180^\circ</math>.<br/> <math>\angle A + \angle QNR = 180^\circ</math><br/> From the triangle QNR, <math>r_1 + r_2 + \angle QNR = 180^\circ</math><br/> Comparing these two equations, we get<br/> <math>r_1 + r_2 = A</math> -----(i)</p> <p>The total deviation <math>\delta</math> is the sum of deviations at the two faces,<br/> <math>\delta = (i - r_1) + (e - r_2)</math> that is, <math>\delta = i + e - A</math> -----(ii)</p> <p>When <math>\delta = \delta_m</math>; <math>i = e</math> &amp; <math>r_1 = r_2</math></p> <p>From (i); <math>2r = A</math> or <math>r = A/2</math></p> <p>From (ii); <math>\delta_m = 2i - A</math> or <math>i = \frac{A + \delta_m}{2}</math></p> $\mu = \frac{\sin i}{\sin r} = \frac{\sin(\frac{A + \delta_m}{2})}{\sin \frac{A}{2}}$ <p>(ii) Given; <math>P = -5D</math></p> $f \text{ (in cm)} = \frac{100}{(-5)} = -20 \text{ cm}$ <p>Using Lens Maker's formula ; <math>\frac{1}{f} = (\mu - 1) \left[ \frac{1}{R_1} - \frac{1}{R_2} \right]</math></p> $\frac{1}{(-20)} = (\mu - 1) \left[ \frac{1}{(-20)} - \frac{1}{(+20)} \right]$ $\frac{1}{(-20)} = (\mu - 1) \left[ -\frac{1}{10} \right]; \quad \mu - 1 = \frac{1}{2}$ $\Rightarrow \mu = \frac{3}{2} = 1.5$ | <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> | 5 |
|     | <b>SECTION E</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                 |   |
| 34. | <div style="border: 1px solid black; padding: 5px; margin-bottom: 10px;"> <p>(a) Time taken by the electron to strike the edge. 1</p> <p>(b) Shape of path followed by the electron and it's reason <math>\frac{1}{2} + \frac{1}{2}</math></p> <p>(c) Potential Difference applied 2</p> <p style="text-align: center;">OR</p> <p>(c) Magnitude and Direction of magnetic field 1+1</p> </div> <p>(a) Electron strikes the edge after travelling 3 cm horizontally (along x-axis).</p> $S_x = v_x \times t$ $3 \times 10^{-2} = (3 \times 10^7) \times t$ $t = 10^{-9} \text{ s}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p>                                                                                                                                                                                                 |   |

|                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                        |   |                                 |   |                                                               |   |    |  |        |   |                |  |
|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|---|---------------------------------|---|---------------------------------------------------------------|---|----|--|--------|---|----------------|--|
|                                                               | <p>(b) Shape of the path is parabola.<br/>Reason: Force/acceleration is in a fixed direction perpendicular to the initial velocity.</p> <p>(c) Along y-direction</p> $S_y = u_y t + \frac{1}{2} a_y t^2$ $-0.5 \times 10^{-2} = 0 + \frac{1}{2} a_y (10^{-9})^2$ $a_y = -10^{16} \text{ m/s}^2$ <p>Magnitude of acceleration <math>(a_y) = \frac{eE}{m} = \frac{e}{m} \left( \frac{V}{l} \right)</math></p> $V = \frac{10^{16} \times 9.1 \times 10^{-31} \times 1 \times 10^{-2}}{1.6 \times 10^{-19}}$ $V = 568.75 \text{ V}$ <p style="text-align: center;"><b>OR</b></p> <p>(c) <math> qE  =  qvB </math>; <math>B = \frac{E}{v} = \left( \frac{ma_y}{e} \right) \left( \frac{1}{v} \right)</math></p> <p>Along y-direction</p> $S_y = u_y t + \frac{1}{2} a_y t^2$ $-0.5 \times 10^{-2} = 0 + \frac{1}{2} a_y (10^{-9})^2$ $a_y = -10^{16} \text{ m/s}^2$ $B = \left( \frac{9.1 \times 10^{-31} \times 10^{16}}{1.6 \times 10^{-19}} \right) \times \left( \frac{1}{3 \times 10^7} \right)$ $B = 1.9 \times 10^{-3} \text{ T}$ <p>Direction of magnetic field will be out of the plane of the paper.</p> | <p>½<br/>½</p> <p>½</p> <p>½</p> <p>½</p> <p>½</p> <p>½</p> <p>½</p> <p>½</p> <p>½</p> | 4 |                                 |   |                                                               |   |    |  |        |   |                |  |
| 35.                                                           | <table border="1"> <tr> <td>(a) Effect on width of central maximum</td> <td>1</td> </tr> <tr> <td>(b) Condition for first minimum</td> <td>1</td> </tr> <tr> <td>(c) Differences between interference and diffraction patterns</td> <td>2</td> </tr> <tr> <td>OR</td> <td></td> </tr> <tr> <td>Reason</td> <td>2</td> </tr> </table> <p>(a) <math>\beta_o \propto \lambda</math><br/><math>\beta_o</math> will increase with increase in wavelength.</p> <p>(b) When path difference <math>a\theta = \lambda</math> or at an angle ; <math>\theta \approx \lambda/a</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | (a) Effect on width of central maximum                                                 | 1 | (b) Condition for first minimum | 1 | (c) Differences between interference and diffraction patterns | 2 | OR |  | Reason | 2 | <p>1<br/>1</p> |  |
| (a) Effect on width of central maximum                        | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                        |   |                                 |   |                                                               |   |    |  |        |   |                |  |
| (b) Condition for first minimum                               | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                        |   |                                 |   |                                                               |   |    |  |        |   |                |  |
| (c) Differences between interference and diffraction patterns | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                        |   |                                 |   |                                                               |   |    |  |        |   |                |  |
| OR                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                        |   |                                 |   |                                                               |   |    |  |        |   |                |  |
| Reason                                                        | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                        |   |                                 |   |                                                               |   |    |  |        |   |                |  |

(c) Differences

| Interference Pattern                                                                                                                                                                                                                                                                                                                                                             | Diffraction Pattern                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>(i) All the maxima are equally spaced.</p> <p>(ii) The dark fringe is having zero intensity.</p> <p>(iii) All the maxima are of the same intensity.</p> <p>(iv)</p>  <p><b>Alternatively:-</b><br/>-It is obtained by the superposition of two waves originating from two sources/slits.</p> | <p>(i) Width of Central bright maxima is twice the width of the other maxima.</p> <p>(ii) The dark fringe is not completely dark.</p> <p>(iii) There is a sharp decrease in the intensity of maxima after the central bright maxima.</p> <p>(iv)</p>  <p>It is obtained by the superposition of waves from points on a single slit.</p> |

1+1

Any two of the above differences.

**OR**

(c) The opening (slit) is 3m; which is of the order of the wavelength of sound waves whereas it is very large compare to the wavelength of light.  
Hence, sound can bend around the obstacle while light cannot.

2

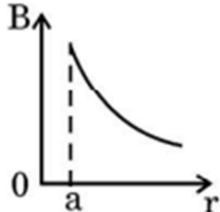
4

**Marking Scheme**  
**Strictly Confidential**  
**(For Internal and Restricted use only)**  
**Senior School Certificate Examination, 2023**  
**SUBJECT NAME PHYSICS (PAPER CODE 55/3/2)**

**General Instructions: -**

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b>  | You are aware that evaluation is the most important process in the actual and correct assessment of the candidates. A small mistake in evaluation may lead to serious problems which may affect the future of the candidates, education system and teaching profession. To avoid mistakes, it is requested that before starting evaluation, you must read and understand the spot evaluation guidelines carefully.                                                                                                                                                                                                                                                                         |
| <b>2</b>  | <b>“Evaluation policy is a confidential policy as it is related to the confidentiality of the examinations conducted, Evaluation done and several other aspects. Its’ leakage to public in any manner could lead to derailment of the examination system and affect the life and future of millions of candidates. Sharing this policy/document to anyone, publishing in any magazine and printing in News Paper/Website etc may invite action under various rules of the Board and IPC.”</b>                                                                                                                                                                                              |
| <b>3</b>  | Evaluation is to be done as per instructions provided in the Marking Scheme. It should not be done according to one’s own interpretation or any other consideration. Marking Scheme should be strictly adhered to and religiously followed. <b>However, while evaluating, answers which are based on latest information or knowledge and/or are innovative, they may be assessed for their correctness otherwise and due marks be awarded to them. In class-X, while evaluating two competency-based questions, please try to understand given answer and even if reply is not from marking scheme but correct competency is enumerated by the candidate, due marks should be awarded.</b> |
| <b>4</b>  | The Marking scheme carries only suggested value points for the answers. These are in the nature of Guidelines only and do not constitute the complete answer. The students can have their own expression and if the expression is correct, the due marks should be awarded accordingly.                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>5</b>  | The Head-Examiner must go through the first five answer books evaluated by each evaluator on the first day, to ensure that evaluation has been carried out as per the instructions given in the Marking Scheme. If there is any variation, the same should be zero after deliberation and discussion. The remaining answer books meant for evaluation shall be given only after ensuring that there is no significant variation in the marking of individual evaluators.                                                                                                                                                                                                                   |
| <b>6</b>  | Evaluators will mark( $\sqrt{\phantom{x}}$ ) wherever answer is correct. For wrong answer CROSS ‘X” be marked. Evaluators will not put right (✓) while evaluating which gives an impression that answer is correct and no marks are awarded. <b>This is most common mistake which evaluators are committing.</b>                                                                                                                                                                                                                                                                                                                                                                           |
| <b>7</b>  | If a question has parts, please award marks on the right-hand side for each part. Marks awarded for different parts of the question should then be totaled up and written in the left-hand margin and encircled. This may be followed strictly.                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>8</b>  | If a question does not have any parts, marks must be awarded in the left-hand margin and encircled. This may also be followed strictly.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>9</b>  | If a student has attempted an extra question, answer of the question deserving more marks should be retained and the other answer scored out with a note <b>“Extra Question”</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>10</b> | No marks to be deducted for the cumulative effect of an error. It should be penalized only once.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>11</b> | A full scale of marks _0-70(example 0 to 80/70/60/50/40/30 marks as given in Question Paper) has to be used. Please do not hesitate to award full marks if the answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           | deserves it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>12</b> | Every examiner has to necessarily do evaluation work for full working hours i.e., 8 hours every day and evaluate 20 answer books per day in main subjects and 25 answer books per day in other subjects (Details are given in Spot Guidelines). This is in view of the reduced syllabus and number of questions in question paper.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>13</b> | <p>Ensure that you do not make the following common types of errors committed by the Examiner in the past:-</p> <ul style="list-style-type: none"> <li>• Leaving answer or part thereof unassessed in an answer book.</li> <li>• Giving more marks for an answer than assigned to it.</li> <li>• Wrong totaling of marks awarded on an answer.</li> <li>• Wrong transfer of marks from the inside pages of the answer book to the title page.</li> <li>• Wrong question wise totaling on the title page.</li> <li>• Wrong totaling of marks of the two columns on the title page.</li> <li>• Wrong grand total.</li> <li>• Marks in words and figures not tallying/not same.</li> <li>• Wrong transfer of marks from the answer book to online award list.</li> <li>• Answers marked as correct, but marks not awarded. (Ensure that the right tick mark is correctly and clearly indicated. It should merely be a line. Same is with the X for incorrect answer.)</li> <li>• Half or a part of answer marked correct and the rest as wrong, but no marks awarded.</li> </ul> |
| <b>14</b> | While evaluating the answer books if the answer is found to be totally incorrect, it should be marked as cross (X) and awarded zero (0) Marks.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>15</b> | Any un assessed portion, non-carrying over of marks to the title page, or totaling error detected by the candidate shall damage the prestige of all the personnel engaged in the evaluation work as also of the Board. Hence, in order to uphold the prestige of all concerned, it is again reiterated that the instructions be followed meticulously and judiciously.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>16</b> | The Examiners should acquaint themselves with the guidelines given in the “ <b>Guidelines for spot Evaluation</b> ” before starting the actual evaluation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>17</b> | Every Examiner shall also ensure that all the answers are evaluated, marks carried over to the title page, correctly totaled and written in figures and words.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>18</b> | The candidates are entitled to obtain photocopy of the Answer Book on request on payment of the prescribed processing fee. All Examiners/Additional Head Examiners/Head Examiners are once again reminded that they must ensure that evaluation is carried out strictly as per value points for each answer as given in the Marking Scheme.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

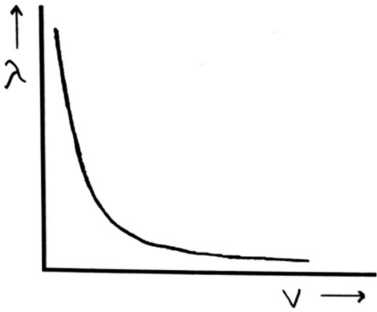
| MARKING SCHEME: PHYSICS(042) |                                                                                                                                                                                                                                                                                                                                                             |       |             |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------|
| Code: 55/3/2                 |                                                                                                                                                                                                                                                                                                                                                             |       |             |
| Q.No.                        | VALUE POINTS/EXPECTED ANSWERS                                                                                                                                                                                                                                                                                                                               | Marks | Total Marks |
| SECTION A                    |                                                                                                                                                                                                                                                                                                                                                             |       |             |
| 1.                           | (a) $\frac{V_o}{\sqrt{2}}$                                                                                                                                                                                                                                                                                                                                  | 1     | 1           |
| 2.                           | (b) $\frac{9}{5}$                                                                                                                                                                                                                                                                                                                                           | 1     | 1           |
| 3.                           | (c) $5 \times 10^9 \text{ m}^{-3}$                                                                                                                                                                                                                                                                                                                          | 1     | 1           |
| 4.                           | (d) $1.0 \times 10^{-7} \text{ Cm}^{-1}$                                                                                                                                                                                                                                                                                                                    | 1     | 1           |
| 5.                           | (d) Infinite                                                                                                                                                                                                                                                                                                                                                | 1     | 1           |
| 6.                           | (b) 0.51 eV                                                                                                                                                                                                                                                                                                                                                 | 1     | 1           |
| 7.                           | (a) Increase                                                                                                                                                                                                                                                                                                                                                | 1     | 1           |
| 8.                           | (b) $\text{m}^2\text{V}^{-1}\text{s}^{-1}$                                                                                                                                                                                                                                                                                                                  | 1     | 1           |
| 9.                           | (b) inductor decreases and the capacitor increases.                                                                                                                                                                                                                                                                                                         | 1     | 1           |
| 10.                          | (c) $\tau \propto I$                                                                                                                                                                                                                                                                                                                                        | 1     | 1           |
| 11.                          | (c) $I \propto A^2$                                                                                                                                                                                                                                                                                                                                         | 1     | 1           |
| 12.                          | (c) H.R. Hertz                                                                                                                                                                                                                                                                                                                                              | 1     | 1           |
| 13.                          | (c) $\frac{2}{3}E$                                                                                                                                                                                                                                                                                                                                          | 1     | 1           |
| 14.                          | (b) twice                                                                                                                                                                                                                                                                                                                                                   | 1     | 1           |
| 15.                          | (c) <div style="text-align: center;">  </div>                                                                                                                                                                                                                            | 1     | 1           |
| 16.                          | (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion(A)                                                                                                                                                                                                                                                | 1     | 1           |
| 17.                          | (c) Assertion (A) is true, but Reason (R) is false.                                                                                                                                                                                                                                                                                                         | 1     | 1           |
| 18.                          | (d) Assertion (A) is false & Reason(R) is also false.                                                                                                                                                                                                                                                                                                       | 1     | 1           |
| SECTION B                    |                                                                                                                                                                                                                                                                                                                                                             |       |             |
| 19.                          | <div style="border: 1px solid black; padding: 5px;">           Effect on width of depletion layer with justification in case of<br/>           (i) Forward Bias <span style="float: right;"><math>\frac{1}{2} + \frac{1}{2}</math></span><br/>           (ii) Reverse Bias <span style="float: right;"><math>\frac{1}{2} + \frac{1}{2}</math></span> </div> |       |             |

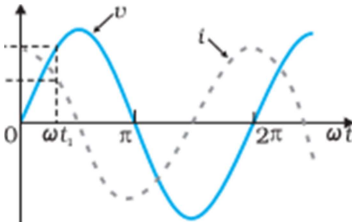


|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                              |   |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|     | <p style="text-align: center;"><b>OR</b></p> <p>(b)</p> <div style="border: 1px solid black; padding: 5px; margin: 10px 0;">             Expression for radius of the <math>n^{\text{th}}</math> orbit in a hydrogen atom <span style="float: right;">2</span> </div> $\frac{mv^2}{r_n} = \frac{kq^2}{r_n^2} = \frac{e^2}{4\pi\epsilon_0 r_n^2} \quad \text{-----(1)}$ $mvr_n = \frac{nh}{2\pi} \quad \text{-----(2)}$ <p>Using equation (1) &amp; (2)</p> $r_n = \frac{n^2 h^2 4\pi\epsilon_0}{m(2\pi)^2 e^2} = 0.53 \times 10^{-10} n^2 \text{ m}$                                                                                                                                                                                                                                                 | <p style="text-align: center;"><math>\frac{1}{2}</math></p> <p style="text-align: center;"><math>\frac{1}{2}</math></p> <p style="text-align: center;">1</p> | 2 |
| 22. | <div style="border: 1px solid black; padding: 5px; margin: 10px 0;">             Definition of Displacement Current <span style="float: right;">1</span><br/>             Difference <span style="float: right;">1</span> </div> <p>Displacement current: It is the current that arises due to the rate of change of electric field/flux.</p> <p><b>Alternatively:-</b></p> $I_d = \epsilon_0 \left( \frac{d\phi_E}{dt} \right)$ <p><b>Alternatively:</b> The term with units of current to explain the continuity of current in a region.</p> <p><b>Difference:</b></p> <p>Displacement current is due to change in electric flux.<br/> Conduction current is due to flow of electrons.</p> <p><b>Alternatively:</b></p> $I_d = \epsilon_0 \left( \frac{d\phi_E}{dt} \right)$ $I_c = \frac{dq}{dt}$ | <p style="text-align: center;">1</p> <p style="text-align: center;">1</p>                                                                                    | 2 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                              |   |

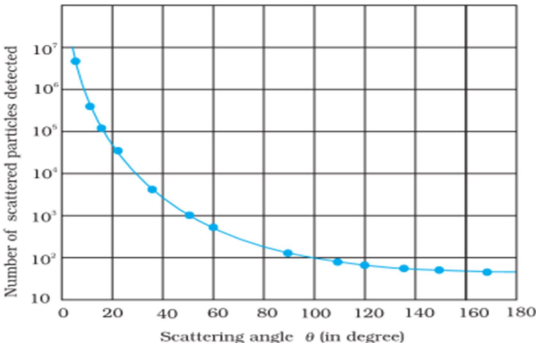


|                               |                                                                                                                                                                                                                                                                                                                                                                                                 |                               |                            |                                                                                                                                                 |      |                         |   |  |  |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|------|-------------------------|---|--|--|
|                               | <div>OR</div> <div>(b)</div> <table><tr><td>(i)</td><td>Diagram for concave mirror</td><td>1</td></tr><tr><td>(ii)</td><td>Diagram for convex lens</td><td>1</td></tr></table>                                                                                                                                                                                                                  | (i)                           | Diagram for concave mirror | 1                                                                                                                                               | (ii) | Diagram for convex lens | 1 |  |  |
| (i)                           | Diagram for concave mirror                                                                                                                                                                                                                                                                                                                                                                      | 1                             |                            |                                                                                                                                                 |      |                         |   |  |  |
| (ii)                          | Diagram for convex lens                                                                                                                                                                                                                                                                                                                                                                         | 1                             |                            |                                                                                                                                                 |      |                         |   |  |  |
|                               | <div>(i)</div> <div></div> <div>(ii)</div> <div></div>                                                                                                                                                                                                                                                                                                                                          | 1                             | 1                          |                                                                                                                                                 |      |                         |   |  |  |
|                               |                                                                                                                                                                                                                                                                                                                                                                                                 |                               | 2                          |                                                                                                                                                 |      |                         |   |  |  |
| 25.                           | <table><tr><td>Calculation of critical angle</td><td>2</td></tr></table> <div>From Snell 's law:-</div> <div><math display="block">\mu_A \sin i_c = \mu_B \sin 90^\circ</math></div> <div><math display="block">2 \times \sin i_c = \sqrt{2} \times 1</math></div> <div><math display="block">\sin i_c = \frac{1}{\sqrt{2}}</math></div> <div><math display="block">i_c = 45^\circ</math></div> | Calculation of critical angle | 2                          | <div><math>\frac{1}{2}</math></div> <div><math>\frac{1}{2}</math></div> <div><math>\frac{1}{2}</math></div> <div><math>\frac{1}{2}</math></div> |      |                         |   |  |  |
| Calculation of critical angle | 2                                                                                                                                                                                                                                                                                                                                                                                               |                               |                            |                                                                                                                                                 |      |                         |   |  |  |

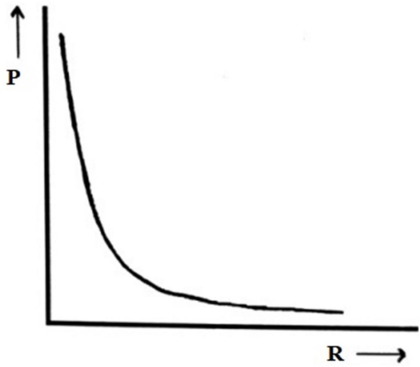
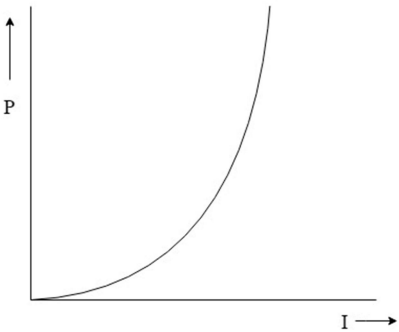
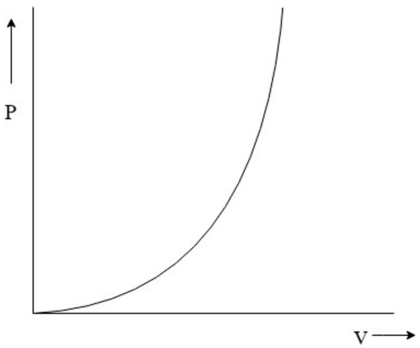
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                  |   |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---|
|     | <p><b>Alternatively:</b></p> $\sin i_c = \frac{1}{\mu_A}$ $\sin i_c = \frac{1}{\sqrt{2}}$ $i_c = 45^\circ$                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1<br>½<br>½      | 2 |
|     | <b>SECTION C</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |   |
| 26. | <div> <div>(a) Graph 1</div> <div>(b) Calculation of energy &amp; de-Broglie Wavelength 1+ 1</div> </div> <p>(a)</p>  <p>(b) Energy acquired = <math>qV_o</math></p> $= 1.6 \times 10^{-19} \times 400$ $= 6.4 \times 10^{-17} \text{ J} = 400 \text{ eV}$ $\lambda = \frac{h}{\sqrt{2mK}}$ $\lambda = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 6.4 \times 10^{-17}}}$ $\lambda = 0.64 \times 10^{-10} \text{ m} = 0.6145 \text{ Å}$ | 1<br>½<br>½<br>½ | 3 |
| 27. | <p>(a)</p> <div> <div>(i) Expression for current 1</div> <div>(ii) Reactance of the capacitor 1</div> <div>Graph of i versus ωt 1</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                      |                  |   |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                       |   |                  |  |                              |               |                               |               |                                                                                                                                                                                   |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|---|------------------|--|------------------------------|---------------|-------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <p>(i) <math>V_m \sin \omega t = \frac{q}{C}</math></p> $I = \frac{dq}{dt} = \frac{d}{dt}(CV_m \sin \omega t)$ $I = \omega CV_m \cos \omega t$ <p><b>Alternatively:-</b></p> $I = \frac{V_m}{\frac{1}{\omega C}} \cos \omega t = I_m \sin(\omega t + \pi/2)$ <p>(ii) <math>I = \frac{V_m}{\frac{1}{\omega C}} \sin(\omega t + \pi/2) = I_m \sin(\omega t + \pi/2)</math></p> <p>Comparing with <math>I_m = \frac{V_m}{\frac{1}{\omega C}}</math></p> <p>Reactance of the capacitor; <math>X_c = \frac{1}{\omega C}</math></p>  <p style="text-align: center;"><b>OR</b></p> <p>(b)</p> <table border="1"><tr><td>Expression for average power consumed</td><td>2</td></tr><tr><td>Power Factor for</td><td></td></tr><tr><td>(i) Purely Inductive circuit</td><td><math>\frac{1}{2}</math></td></tr><tr><td>(ii) Purely Resistive Circuit</td><td><math>\frac{1}{2}</math></td></tr></table> <p>Instantaneous Power;</p> $P = VI = (V_m \sin \omega t) \times i_m \sin(\omega t + \phi)$ $P = \frac{V_m i_m}{2} [\cos \phi - \cos(2\omega t + \phi)] \quad \text{----(1)}$ <p>The average power over a cycle is given by the average of the two terms in the R.H.S of equation (1). It is only the second term which is time dependent .Its average is zero (the positive half of the cosine cancels the negative half).<br/>Therefore,</p> | Expression for average power consumed | 2 | Power Factor for |  | (i) Purely Inductive circuit | $\frac{1}{2}$ | (ii) Purely Resistive Circuit | $\frac{1}{2}$ | <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p>1</p> <p>1</p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> |  |
| Expression for average power consumed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2                                     |   |                  |  |                              |               |                               |               |                                                                                                                                                                                   |  |
| Power Factor for                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                       |   |                  |  |                              |               |                               |               |                                                                                                                                                                                   |  |
| (i) Purely Inductive circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\frac{1}{2}$                         |   |                  |  |                              |               |                               |               |                                                                                                                                                                                   |  |
| (ii) Purely Resistive Circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\frac{1}{2}$                         |   |                  |  |                              |               |                               |               |                                                                                                                                                                                   |  |

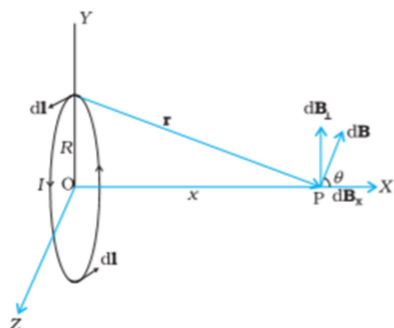


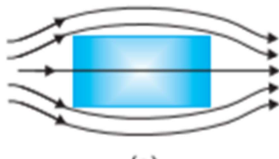
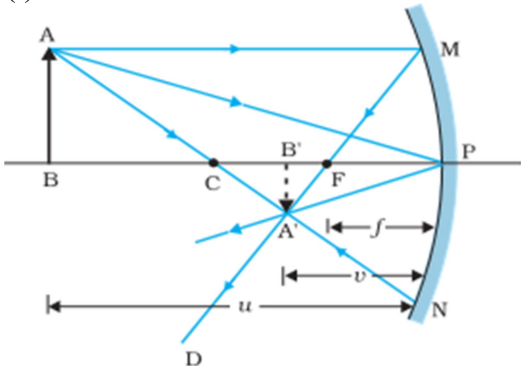
| <p>Inferences</p> <ul style="list-style-type: none"><li>-The force is attractive for distances larger than <math>r_0</math>.</li><li>- The force is repulsive for distance less than <math>r_0</math>.</li></ul> <p><b>Alternatively:-</b></p> <p>Any other relevant inference drawn from the graph should be given full credit.</p> <p style="text-align: center;"><b>OR</b></p> <p>(b)</p> <table border="1"><tr><td>(i) Graph to show the variation of the number of scattered particles as a function of scattering angle.</td><td>1</td></tr><tr><td>(ii) Two conclusions</td><td><math>\frac{1}{2} + \frac{1}{2}</math></td></tr><tr><td>Discovery of nucleus</td><td>1</td></tr></table> <p>(i)</p> <div><table border="1"><caption>Data points estimated from the graph</caption><thead><tr><th>Scattering angle <math>\theta</math> (in degree)</th><th>Number of scattered particles detected</th></tr></thead><tbody><tr><td>0</td><td><math>10^7</math></td></tr><tr><td>10</td><td><math>10^6</math></td></tr><tr><td>20</td><td><math>10^5</math></td></tr><tr><td>30</td><td><math>10^4</math></td></tr><tr><td>40</td><td><math>10^3</math></td></tr><tr><td>60</td><td><math>10^2</math></td></tr><tr><td>90</td><td><math>10^1</math></td></tr><tr><td>120</td><td><math>10^1</math></td></tr><tr><td>150</td><td><math>10^1</math></td></tr><tr><td>180</td><td><math>10^1</math></td></tr></tbody></table></div> <p>(ii)- The entire positive charge and most of the mass of the atom are concentrated in a small space.</p> <p>-Many of the <math>\alpha</math>-particles pass through the foil. It means that they do not suffer any collisions.</p> <p>To deflect the <math>\alpha</math>-particle backwards, a large repulsive force is required, which is provided only if the greater part of the mass of the atom &amp; its positive charge were concentrated tightly at its centre. This lead to the discovery of the nucleus in the atom.</p> | (i) Graph to show the variation of the number of scattered particles as a function of scattering angle. | 1 | (ii) Two conclusions | $\frac{1}{2} + \frac{1}{2}$ | Discovery of nucleus | 1 | Scattering angle $\theta$ (in degree) | Number of scattered particles detected | 0 | $10^7$ | 10 | $10^6$ | 20 | $10^5$ | 30 | $10^4$ | 40 | $10^3$ | 60 | $10^2$ | 90 | $10^1$ | 120 | $10^1$ | 150 | $10^1$ | 180 | $10^1$ | <p><math>\frac{1}{2}</math><br/><math>\frac{1}{2}</math></p> <p>1</p> <p><math>\frac{1}{2}</math><br/><math>\frac{1}{2}</math></p> <p>1</p> | <p>3</p> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---|----------------------|-----------------------------|----------------------|---|---------------------------------------|----------------------------------------|---|--------|----|--------|----|--------|----|--------|----|--------|----|--------|----|--------|-----|--------|-----|--------|-----|--------|---------------------------------------------------------------------------------------------------------------------------------------------|----------|
| (i) Graph to show the variation of the number of scattered particles as a function of scattering angle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1                                                                                                       |   |                      |                             |                      |   |                                       |                                        |   |        |    |        |    |        |    |        |    |        |    |        |    |        |     |        |     |        |     |        |                                                                                                                                             |          |
| (ii) Two conclusions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $\frac{1}{2} + \frac{1}{2}$                                                                             |   |                      |                             |                      |   |                                       |                                        |   |        |    |        |    |        |    |        |    |        |    |        |    |        |     |        |     |        |     |        |                                                                                                                                             |          |
| Discovery of nucleus                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1                                                                                                       |   |                      |                             |                      |   |                                       |                                        |   |        |    |        |    |        |    |        |    |        |    |        |    |        |     |        |     |        |     |        |                                                                                                                                             |          |
| Scattering angle $\theta$ (in degree)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Number of scattered particles detected                                                                  |   |                      |                             |                      |   |                                       |                                        |   |        |    |        |    |        |    |        |    |        |    |        |    |        |     |        |     |        |     |        |                                                                                                                                             |          |
| 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $10^7$                                                                                                  |   |                      |                             |                      |   |                                       |                                        |   |        |    |        |    |        |    |        |    |        |    |        |    |        |     |        |     |        |     |        |                                                                                                                                             |          |
| 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $10^6$                                                                                                  |   |                      |                             |                      |   |                                       |                                        |   |        |    |        |    |        |    |        |    |        |    |        |    |        |     |        |     |        |     |        |                                                                                                                                             |          |
| 20                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $10^5$                                                                                                  |   |                      |                             |                      |   |                                       |                                        |   |        |    |        |    |        |    |        |    |        |    |        |    |        |     |        |     |        |     |        |                                                                                                                                             |          |
| 30                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $10^4$                                                                                                  |   |                      |                             |                      |   |                                       |                                        |   |        |    |        |    |        |    |        |    |        |    |        |    |        |     |        |     |        |     |        |                                                                                                                                             |          |
| 40                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $10^3$                                                                                                  |   |                      |                             |                      |   |                                       |                                        |   |        |    |        |    |        |    |        |    |        |    |        |    |        |     |        |     |        |     |        |                                                                                                                                             |          |
| 60                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $10^2$                                                                                                  |   |                      |                             |                      |   |                                       |                                        |   |        |    |        |    |        |    |        |    |        |    |        |    |        |     |        |     |        |     |        |                                                                                                                                             |          |
| 90                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $10^1$                                                                                                  |   |                      |                             |                      |   |                                       |                                        |   |        |    |        |    |        |    |        |    |        |    |        |    |        |     |        |     |        |     |        |                                                                                                                                             |          |
| 120                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $10^1$                                                                                                  |   |                      |                             |                      |   |                                       |                                        |   |        |    |        |    |        |    |        |    |        |    |        |    |        |     |        |     |        |     |        |                                                                                                                                             |          |
| 150                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $10^1$                                                                                                  |   |                      |                             |                      |   |                                       |                                        |   |        |    |        |    |        |    |        |    |        |    |        |    |        |     |        |     |        |     |        |                                                                                                                                             |          |
| 180                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $10^1$                                                                                                  |   |                      |                             |                      |   |                                       |                                        |   |        |    |        |    |        |    |        |    |        |    |        |    |        |     |        |     |        |     |        |                                                                                                                                             |          |



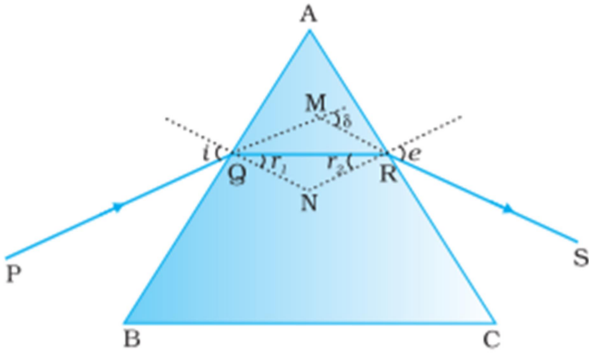
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                               |                            |          |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------|
|  | <p>(a) <math>P = \frac{V^2}{R}</math> ; keeping V constant</p>  <p>(b) <math>P = I^2 R</math> ; keeping R constant</p>  <p>(c) <math>P = \frac{V^2}{R}</math> ; keeping R constant</p>  | <p>1</p> <p>1</p> <p>1</p> | <p>3</p> |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------|

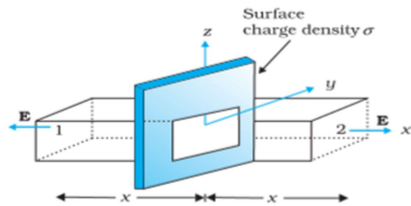
| SECTION D |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |  |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--|
| 31.       | <div>(a)</div> <div><div><div>(i) For a moving coil galvanometer</div><div>Principle1</div><div>Working1</div><div>Reason it cannot be used as such1</div><div>(ii) Reason for radial field1</div><div>How radial field is achieved1</div></div><div>(i) Principle – When a rectangular loop carrying current I is placed in a uniform magnetic field, it experiences a torque.</div><div>Working:-<br/>When a current flows through the coil of a galvanometer, a torque acts on it.<br/><math>\tau = NiAB \sin \theta</math><br/>For radial magnetic field; <math>\sin \theta = 1</math><br/>The spring provides a counter or restoring torque <math>k\phi</math>.<br/><math>k\phi = NiAB</math><br/><b>In equilibrium;</b> <math>\phi = \left(\frac{NAB}{k}\right)i</math></div><div>Galvanometer cannot be used as such to measure current because:<br/>-It has large resistance and hence will change the value of current in the circuit.<br/>-It is a sensitive device.<br/>(Any one of the above)</div><div>(ii) The magnetic field is made radial in a moving coil galvanometer so that the magnetic dipole moment (<math>\vec{m}</math>) is always perpendicular to the magnetic field (<math>\vec{B}</math>) Hence, <math>\sin \theta = 1</math> always<br/><b>Alternatively:</b> The magnetic field is made radial in a moving coil galvanometer to make the scale linear.</div><div>It is achieved by using curved magnetic poles.<br/><b>Alternatively:-</b>By using soft iron cylindrical core.</div><div>OR</div><div>(b)</div><div><div><div>(i) Derivation of expression for magnetic field on the axis of a current carrying loop.</div><div>3</div><div>(ii) Two differences between diamagnetic and paramagnetic substance.</div><div>1+1</div></div></div></div> | 1 |  |

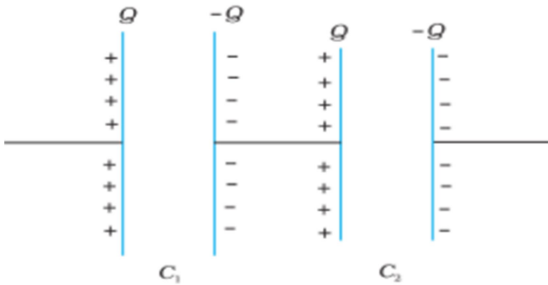
| <p>(i)</p>  $dB = \frac{\mu_0}{4\pi} \frac{I  d\mathbf{l} \times \mathbf{r} }{r^3}$ $= \frac{\mu_0 i dl \sin 90^\circ}{4\pi (x^2 + R^2)^{3/2}}$ <p><math>dB_\perp</math> cancels out.</p> <p>Net B = <math>\int dB_x = \int dB \cos \theta</math></p> $= \frac{\mu_0}{4\pi} \int \frac{idl}{(x^2 + R^2)^{3/2}} \times \frac{R}{(x^2 + R^2)^{1/2}}$ $= \frac{\mu_0 i R}{4\pi (x^2 + R^2)^{3/2}} \int dl$ $= \frac{\mu_0 i R}{4\pi (x^2 + R^2)^{3/2}} (2\pi R)$ $\mathbf{B} = B_x \hat{\mathbf{i}} = \frac{\mu_0 I R^2}{2(x^2 + R^2)^{3/2}} \hat{\mathbf{i}}$ <p>(ii) Differences</p> <table><thead><tr><th>Diamagnetic Materials</th><th>Paramagnetic Materials</th></tr></thead><tbody><tr><td>(i) Susceptibility is between -1 and 0.</td><td>(i) Susceptibility is a small positive number.(slightly greater than zero.)</td></tr><tr><td>(ii) Relative permeability is between 0 and 1.</td><td>(ii) Relative permeability is slightly greater than 1.</td></tr><tr><td>(iii) <math>\mu &lt; \mu_0</math></td><td>(iii) <math>\mu &gt; \mu_0</math></td></tr><tr><td>(iv) Tendency to move from stronger to weaker part of external magnetism.</td><td>(iv) Tendency to move from region of weak to strong magnetic field.</td></tr></tbody></table> | Diamagnetic Materials                                                       | Paramagnetic Materials | (i) Susceptibility is between -1 and 0. | (i) Susceptibility is a small positive number.(slightly greater than zero.) | (ii) Relative permeability is between 0 and 1. | (ii) Relative permeability is slightly greater than 1. | (iii) $\mu < \mu_0$ | (iii) $\mu > \mu_0$ | (iv) Tendency to move from stronger to weaker part of external magnetism. | (iv) Tendency to move from region of weak to strong magnetic field. | <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p>1+1</p> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|------------------------|-----------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------|--------------------------------------------------------|---------------------|---------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Diamagnetic Materials                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Paramagnetic Materials                                                      |                        |                                         |                                                                             |                                                |                                                        |                     |                     |                                                                           |                                                                     |                                                                                                                                                                                                                                            |
| (i) Susceptibility is between -1 and 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | (i) Susceptibility is a small positive number.(slightly greater than zero.) |                        |                                         |                                                                             |                                                |                                                        |                     |                     |                                                                           |                                                                     |                                                                                                                                                                                                                                            |
| (ii) Relative permeability is between 0 and 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | (ii) Relative permeability is slightly greater than 1.                      |                        |                                         |                                                                             |                                                |                                                        |                     |                     |                                                                           |                                                                     |                                                                                                                                                                                                                                            |
| (iii) $\mu < \mu_0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | (iii) $\mu > \mu_0$                                                         |                        |                                         |                                                                             |                                                |                                                        |                     |                     |                                                                           |                                                                     |                                                                                                                                                                                                                                            |
| (iv) Tendency to move from stronger to weaker part of external magnetism.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | (iv) Tendency to move from region of weak to strong magnetic field.         |                        |                                         |                                                                             |                                                |                                                        |                     |                     |                                                                           |                                                                     |                                                                                                                                                                                                                                            |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |  |     |  |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|-----|--|
|     | <div> <div> (v) is repelled by a magnet.<br/>(vi) Field inside the material is reduced.<br/><br/>(vii)<br/>  </div> <div> (v) is weakly attracted by a magnet.<br/>(vi) Field inside is slightly enhanced.<br/><br/>(vii)<br/>  </div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                          |  |  |     |  |
|     | Any two of the above mentioned differences.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |  | 5   |  |
| 32. | <div> (a) <div> <div> (i) Ray diagram showing formation of real image in a concave mirror. 1<br/> Obtaining the relation between u,v and R 2<br/> (ii) Position of image formed 1<br/> Height of image formed 1 </div> <div> (i)  <p>From Fig. the two right-angled triangles A'B'F and MPF are similar. (For paraxial rays, MP can be considered to be a straight line perpendicular to CP.)<br/>Therefore,</p> <math display="block">\frac{B'A'}{PM} = \frac{B'F}{FP}</math> <math display="block">\text{or } \frac{B'A'}{BA} = \frac{B'F}{FP} \quad (\because PM = AB) \quad \text{-----(i)}</math> <p>Since <math>\angle APB = \angle A'PB'</math>, the right angled triangles A'B'P and ABP are also similar. Therefore,</p> </div> </div> </div> |  |  | 1   |  |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |  | 1/2 |  |

|                                                                                                                                                                                                                                                                                                                                                                          |                                                                                         |   |                                                      |   |                                                        |   |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|---|------------------------------------------------------|---|--------------------------------------------------------|---|--|
| $\frac{B'A'}{BA} = \frac{B'P}{BP}$ <p>----- (ii)</p>                                                                                                                                                                                                                                                                                                                     | 1/2                                                                                     |   |                                                      |   |                                                        |   |  |
| <p>Comparing equations (i) and (ii)</p> $\frac{B'F}{FP} = \frac{B'P - FP}{FP} = \frac{B'P}{BP}$ <p>----- (iii)</p>                                                                                                                                                                                                                                                       | 1/2                                                                                     |   |                                                      |   |                                                        |   |  |
| <p><math>B'P = -v, FP = -f, BP = -u;</math></p>                                                                                                                                                                                                                                                                                                                          |                                                                                         |   |                                                      |   |                                                        |   |  |
| <p>Using these in Eq.(iii) we get <math>\frac{1}{v} + \frac{1}{u} = \frac{1}{f} = \frac{2}{R}</math></p>                                                                                                                                                                                                                                                                 | 1/2                                                                                     |   |                                                      |   |                                                        |   |  |
| <p><b>Alternatively:-</b> If the result derived by any other method, full credit to be given.</p>                                                                                                                                                                                                                                                                        |                                                                                         |   |                                                      |   |                                                        |   |  |
| <p>(ii) For lens: <math>\frac{1}{v} - \frac{1}{u} = \frac{1}{f}</math></p> <p><math>u = -5m; \quad f = +1m</math></p> $\frac{1}{v} - \frac{1}{-5} = \frac{1}{(+1)}$ $\Rightarrow v = \frac{5}{4}m = 1.25m$ $m = \frac{I}{O} = \frac{v}{u} = \frac{\left(+\frac{5}{4}\right)}{(-5)}$ <p><math>I = (-0.25) \times (1.8)</math></p> <p><math>I = -0.45 \text{ m}</math></p> | 1/2                                                                                     |   |                                                      |   |                                                        |   |  |
| <p style="text-align: center;"><b>OR</b></p>                                                                                                                                                                                                                                                                                                                             |                                                                                         |   |                                                      |   |                                                        |   |  |
| <p>(b)</p>                                                                                                                                                                                                                                                                                                                                                               |                                                                                         |   |                                                      |   |                                                        |   |  |
| <table> <tr> <td>(i) Ray diagram showing refraction of a ray of light through a rectangular glass prism.</td> <td>1</td> </tr> <tr> <td>Obtaining the relation between <math>\mu, A</math> &amp; <math>\delta_m</math></td> <td>2</td> </tr> <tr> <td>(ii) Finding Refractive Index of material of the lens.</td> <td>2</td> </tr> </table>                              | (i) Ray diagram showing refraction of a ray of light through a rectangular glass prism. | 1 | Obtaining the relation between $\mu, A$ & $\delta_m$ | 2 | (ii) Finding Refractive Index of material of the lens. | 2 |  |
| (i) Ray diagram showing refraction of a ray of light through a rectangular glass prism.                                                                                                                                                                                                                                                                                  | 1                                                                                       |   |                                                      |   |                                                        |   |  |
| Obtaining the relation between $\mu, A$ & $\delta_m$                                                                                                                                                                                                                                                                                                                     | 2                                                                                       |   |                                                      |   |                                                        |   |  |
| (ii) Finding Refractive Index of material of the lens.                                                                                                                                                                                                                                                                                                                   | 2                                                                                       |   |                                                      |   |                                                        |   |  |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                          |          |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|  | <p>(i)</p>  <p>In the quadrilateral AQNR, two of the angles (at the vertices Q and R) are right angles. Therefore, the sum of the other angles of the quadrilateral is <math>180^\circ</math>.<br/> <math>\angle A + \angle QNR = 180^\circ</math><br/> From the triangle QNR, <math>r_1 + r_2 + \angle QNR = 180^\circ</math><br/> Comparing these two equations, we get<br/> <math>r_1 + r_2 = A</math> -----(i)<br/> The total deviation <math>\delta</math> is the sum of deviations at the two faces,<br/> <math>\delta = (i - r_1) + (e - r_2)</math> that is, <math>\delta = i + e - A</math> -----(ii)</p> <p>When <math>\delta = \delta_m</math>; <math>i = e</math> &amp; <math>r_1 = r_2</math><br/> From (i); <math>2r = A</math> or <math>r = A/2</math><br/> From (ii); <math>\delta_m = 2i - A</math> or <math>i = \frac{A + \delta_m}{2}</math></p> $\mu = \frac{\sin i}{\sin r} = \frac{\sin(\frac{A + \delta_m}{2})}{\sin \frac{A}{2}}$ <p>(ii) Given; <math>P = -5D</math><br/> <math>f \text{ (in cm)} = \frac{100}{(-5)} = -20 \text{ cm}</math></p> <p>Using Lens Maker's formula ; <math>\frac{1}{f} = (\mu - 1) \left[ \frac{1}{R_1} - \frac{1}{R_2} \right]</math></p> $\frac{1}{(-20)} = (\mu - 1) \left[ \frac{1}{(-20)} - \frac{1}{(+20)} \right]$ $\frac{1}{(-20)} = (\mu - 1) \left[ -\frac{1}{10} \right]; \quad \mu - 1 = \frac{1}{2}$ $\Rightarrow \mu = \frac{3}{2} = 1.5$ | <p>1</p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> | <p>5</p> |
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|                                                                                                  |                                                                                                                                                                                                                                                                                                                                                               |                                           |                             |                                                                                                  |   |                                       |   |  |  |
|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-----------------------------|--------------------------------------------------------------------------------------------------|---|---------------------------------------|---|--|--|
| 33.                                                                                              | <p>(a)</p> <table border="1"> <tr> <td>(i) Definition &amp; SI Unit of Electric Flux</td> <td><math>\frac{1}{2} + \frac{1}{2}</math></td> </tr> <tr> <td>(ii) Deriving the expression for electric field due to a uniformly charged infinite plane sheet.</td> <td>2</td> </tr> <tr> <td>(iii) Net charge enclosed by the cube</td> <td>2</td> </tr> </table> | (i) Definition & SI Unit of Electric Flux | $\frac{1}{2} + \frac{1}{2}$ | (ii) Deriving the expression for electric field due to a uniformly charged infinite plane sheet. | 2 | (iii) Net charge enclosed by the cube | 2 |  |  |
| (i) Definition & SI Unit of Electric Flux                                                        | $\frac{1}{2} + \frac{1}{2}$                                                                                                                                                                                                                                                                                                                                   |                                           |                             |                                                                                                  |   |                                       |   |  |  |
| (ii) Deriving the expression for electric field due to a uniformly charged infinite plane sheet. | 2                                                                                                                                                                                                                                                                                                                                                             |                                           |                             |                                                                                                  |   |                                       |   |  |  |
| (iii) Net charge enclosed by the cube                                                            | 2                                                                                                                                                                                                                                                                                                                                                             |                                           |                             |                                                                                                  |   |                                       |   |  |  |
|                                                                                                  | <p>(i) <math>\phi = \vec{E} \cdot \vec{A}</math></p> <p><b>Alternatively:</b> Electric flux is the number of electric field lines passing through an area normally.<br/>S.I. unit of electric flux <math>\text{Nm}^2/\text{C}</math> or <math>\text{V-m}</math>.</p>                                                                                          | $\frac{1}{2}$                             | $\frac{1}{2}$               |                                                                                                  |   |                                       |   |  |  |
|                                                                                                  | <p>(ii)</p>                                                                                                                                                                                                                                                                  | $\frac{1}{2}$                             |                             |                                                                                                  |   |                                       |   |  |  |
|                                                                                                  | <p>From Gauss's law:- <math>\phi = \oint \vec{E} \cdot d\vec{A} = \frac{q}{\epsilon_0}</math></p> $2EA = \frac{\sigma A}{\epsilon_0}$ $E = \frac{\sigma}{2\epsilon_0}$                                                                                                                                                                                        | $\frac{1}{2}$                             | $\frac{1}{2}$               |                                                                                                  |   |                                       |   |  |  |
|                                                                                                  | <p><b>Alternatively:</b> If the shape of the Gaussian surface is taken cylindrical, full credit to be given.</p>                                                                                                                                                                                                                                              | $\frac{1}{2}$                             |                             |                                                                                                  |   |                                       |   |  |  |
|                                                                                                  | <p>(iii)</p> $\phi_L = E ds \cos 180^\circ = -E ds$ $= -BL^2$ $\phi_R = E ds \cos 0^\circ = E ds$ $= (AL + B)L^2 = AL^3 + BL^2$ <p>Net flux <math>= \phi_L + \phi_R</math></p> $= (AL^3 + BL^2) - BL^2$ <p>Net flux <math>= AL^3 = \frac{q}{\epsilon_0}</math></p> $\Rightarrow q = AL^3 \epsilon_0$                                                          | $\frac{1}{2}$                             | $\frac{1}{2}$               |                                                                                                  |   |                                       |   |  |  |
|                                                                                                  |                                                                                                                                                                                                                                                                                                                                                               | $\frac{1}{2}$                             | $\frac{1}{2}$               |                                                                                                  |   |                                       |   |  |  |

| OR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                             |                                                                                                                                                                                                 |                             |                                                         |   |                                                     |   |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|---------------------------------------------------------|---|-----------------------------------------------------|---|--|
| (b)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                             |                                                                                                                                                                                                 |                             |                                                         |   |                                                     |   |  |
| <table border="1"> <tr> <td>(i) Definitions &amp; S.I. Unit of electric potential</td> <td><math>\frac{1}{2} + \frac{1}{2}</math></td> </tr> <tr> <td>(ii) Derivation of expression of Equivalent capacitance</td> <td>2</td> </tr> <tr> <td>(iii) Calculation of Electrostatic Potential Energy</td> <td>2</td> </tr> </table>                                                                                                                                                                                                                                                                                                                                                                                                         |                             | (i) Definitions & S.I. Unit of electric potential                                                                                                                                               | $\frac{1}{2} + \frac{1}{2}$ | (ii) Derivation of expression of Equivalent capacitance | 2 | (iii) Calculation of Electrostatic Potential Energy | 2 |  |
| (i) Definitions & S.I. Unit of electric potential                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\frac{1}{2} + \frac{1}{2}$ |                                                                                                                                                                                                 |                             |                                                         |   |                                                     |   |  |
| (ii) Derivation of expression of Equivalent capacitance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2                           |                                                                                                                                                                                                 |                             |                                                         |   |                                                     |   |  |
| (iii) Calculation of Electrostatic Potential Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2                           |                                                                                                                                                                                                 |                             |                                                         |   |                                                     |   |  |
| <p>(i) Electrical Potential – Electrostatic potential at any point in a region with electrostatic field is the work done in bringing a unit positive charge (without acceleration) from infinity to that point.</p> <p><b>Alternatively:-</b></p> $V = \frac{\text{Work Done}}{q}$ $V = -\int \vec{E} \cdot d\vec{l}$ <p>S.I. unit of electrostatic potential is volt.</p> <p><b>Alternatively:-</b></p> <p>S.I. unit is J/C.</p> <div style="text-align: center;">  </div> $V = V_1 + V_2 = \frac{Q}{C_1} + \frac{Q}{C_2}$ $\frac{Q}{C_{eq.}} = Q \left( \frac{1}{C_1} + \frac{1}{C_2} \right)$ $\frac{1}{C_{eq.}} = \frac{1}{C_1} + \frac{1}{C_2}$ |                             | <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> |                             |                                                         |   |                                                     |   |  |
| <p>(iii)</p> <p>Potential energy of the system = <math>K \left[ \frac{Q(-q)}{4a} + \frac{Qq}{3a} - \frac{q^2}{5a} \right]</math></p> <p>Potential energy of the system = 0</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                             | $\frac{1}{2}$                                                                                                                                                                                   |                             |                                                         |   |                                                     |   |  |



|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                 |   |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|     | <div> <div>two waves originating from two sources/slits.</div> <div>of waves from points on a single slit.</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                 |   |
|     | <p>Any two of the above differences.</p> <p style="text-align: center;"><b>OR</b></p> <p>(c) The opening (slit) is 3m; which is of the order of the wavelength of sound waves whereas it is very large compare to the wavelength of light. Hence, sound can bend around the obstacle while light cannot.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2                                                                                                                                                                                               | 4 |
| 35. | <div> <div> <div>(a) Time taken by the electron to strike the edge. 1</div> <div>(b) Shape of path followed by the electron and it's reason <math>\frac{1}{2} + \frac{1}{2}</math></div> <div>(c) Potential Difference applied 2</div> </div> <div> <div>OR</div> <div>(c) Magnitude and Direction of magnetic field 1+1</div> </div> </div> <p>(a) Electron strikes the edge after travelling 3 cm horizontally (along x-axis).</p> $S_x = v_x \times t$ $3 \times 10^{-2} = (3 \times 10^7) \times t$ $t = 10^{-9} \text{ s}$ <p>(b) Shape of the path is parabola.<br/>Reason: Force/acceleration is in a fixed direction perpendicular to the initial velocity.</p> <p>(c) Along y-direction</p> $S_y = u_y t + \frac{1}{2} a_y t^2$ $-0.5 \times 10^{-2} = 0 + \frac{1}{2} a_y (10^{-9})^2$ $a_y = -10^{16} \text{ m/s}^2$ <p>Magnitude of acceleration <math>(a_y) = \frac{eE}{m} = \frac{e}{m} \left( \frac{V}{l} \right)</math></p> $V = \frac{10^{16} \times 9.1 \times 10^{-31} \times 1 \times 10^{-2}}{1.6 \times 10^{-19}}$ $V = 568.75 \text{ V}$ <p style="text-align: center;"><b>OR</b></p> <p>(c) <math> qE  =  qvB </math>; <math>B = \frac{E}{v} = \left( \frac{ma_y}{e} \right) \left( \frac{1}{v} \right)</math></p> <p>Along y-direction</p> | <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> |   |

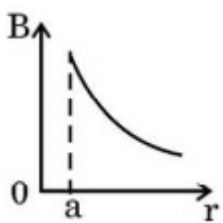


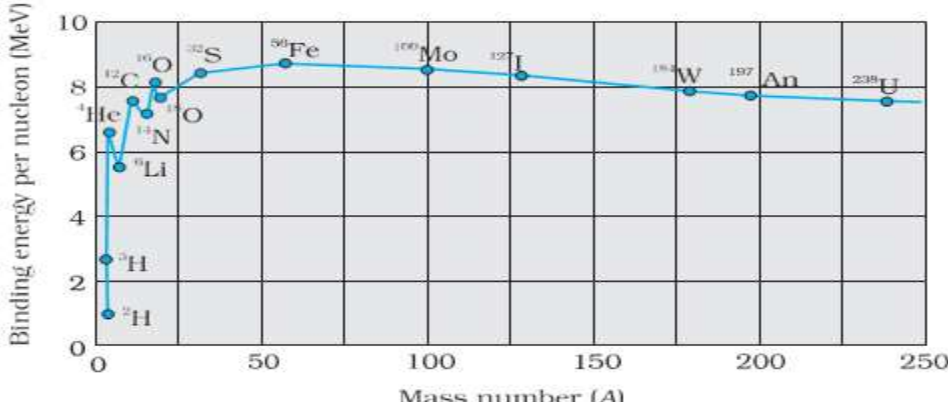
**Marking Scheme**  
**Strictly Confidential**  
**(For Internal and Restricted use only)**  
**Senior School Certificate Examination, 2023**  
**SUBJECT NAME PHYSICS (PAPER CODE 55/3/3)**

**General Instructions: -**

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b>  | You are aware that evaluation is the most important process in the actual and correct assessment of the candidates. A small mistake in evaluation may lead to serious problems which may affect the future of the candidates, education system and teaching profession. To avoid mistakes, it is requested that before starting evaluation, you must read and understand the spot evaluation guidelines carefully.                                                                                                                                                                                                                                                                         |
| <b>2</b>  | <b>“Evaluation policy is a confidential policy as it is related to the confidentiality of the examinations conducted, Evaluation done and several other aspects. Its’ leakage to public in any manner could lead to derailment of the examination system and affect the life and future of millions of candidates. Sharing this policy/document to anyone, publishing in any magazine and printing in News Paper/Website etc may invite action under various rules of the Board and IPC.”</b>                                                                                                                                                                                              |
| <b>3</b>  | Evaluation is to be done as per instructions provided in the Marking Scheme. It should not be done according to one’s own interpretation or any other consideration. Marking Scheme should be strictly adhered to and religiously followed. <b>However, while evaluating, answers which are based on latest information or knowledge and/or are innovative, they may be assessed for their correctness otherwise and due marks be awarded to them. In class-X, while evaluating two competency-based questions, please try to understand given answer and even if reply is not from marking scheme but correct competency is enumerated by the candidate, due marks should be awarded.</b> |
| <b>4</b>  | The Marking scheme carries only suggested value points for the answers. These are in the nature of Guidelines only and do not constitute the complete answer. The students can have their own expression and if the expression is correct, the due marks should be awarded accordingly.                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>5</b>  | The Head-Examiner must go through the first five answer books evaluated by each evaluator on the first day, to ensure that evaluation has been carried out as per the instructions given in the Marking Scheme. If there is any variation, the same should be zero after deliberation and discussion. The remaining answer books meant for evaluation shall be given only after ensuring that there is no significant variation in the marking of individual evaluators.                                                                                                                                                                                                                   |
| <b>6</b>  | Evaluators will mark( $\sqrt{\phantom{x}}$ ) wherever answer is correct. For wrong answer CROSS ‘X” be marked. Evaluators will not put right (✓) while evaluating which gives an impression that answer is correct and no marks are awarded. <b>This is most common mistake which evaluators are committing.</b>                                                                                                                                                                                                                                                                                                                                                                           |
| <b>7</b>  | If a question has parts, please award marks on the right-hand side for each part. Marks awarded for different parts of the question should then be totaled up and written in the left-hand margin and encircled. This may be followed strictly.                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>8</b>  | If a question does not have any parts, marks must be awarded in the left-hand margin and encircled. This may also be followed strictly.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>9</b>  | If a student has attempted an extra question, answer of the question deserving more marks should be retained and the other answer scored out with a note <b>“Extra Question”</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>10</b> | No marks to be deducted for the cumulative effect of an error. It should be penalized only once.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>11</b> | A full scale of marks 0 - 70 (example 0 to 80/70/60/50/40/30 marks as given in Question Paper) has to be used. Please do not hesitate to award full marks if the answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

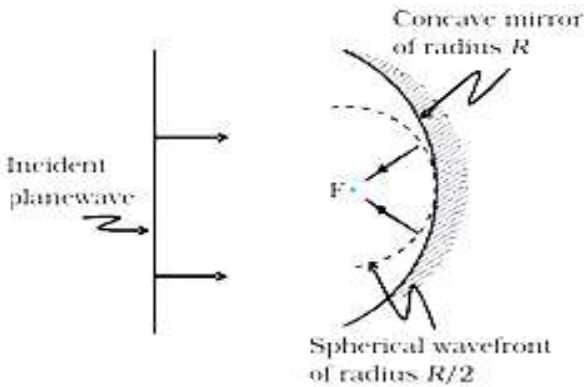
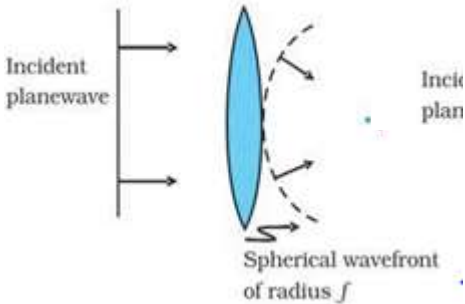
|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           | deserves it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>12</b> | Every examiner has to necessarily do evaluation work for full working hours i.e., 8 hours every day and evaluate 20 answer books per day in main subjects and 25 answer books per day in other subjects (Details are given in Spot Guidelines). This is in view of the reduced syllabus and number of questions in question paper.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>13</b> | <p>Ensure that you do not make the following common types of errors committed by the Examiner in the past:-</p> <ul style="list-style-type: none"> <li>• Leaving answer or part thereof unassessed in an answer book.</li> <li>• Giving more marks for an answer than assigned to it.</li> <li>• Wrong totaling of marks awarded on an answer.</li> <li>• Wrong transfer of marks from the inside pages of the answer book to the title page.</li> <li>• Wrong question wise totaling on the title page.</li> <li>• Wrong totaling of marks of the two columns on the title page.</li> <li>• Wrong grand total.</li> <li>• Marks in words and figures not tallying/not same.</li> <li>• Wrong transfer of marks from the answer book to online award list.</li> <li>• Answers marked as correct, but marks not awarded. (Ensure that the right tick mark is correctly and clearly indicated. It should merely be a line. Same is with the X for incorrect answer.)</li> <li>• Half or a part of answer marked correct and the rest as wrong, but no marks awarded.</li> </ul> |
| <b>14</b> | While evaluating the answer books if the answer is found to be totally incorrect, it should be marked as cross (X) and awarded zero (0) Marks.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>15</b> | Any un assessed portion, non-carrying over of marks to the title page, or totaling error detected by the candidate shall damage the prestige of all the personnel engaged in the evaluation work as also of the Board. Hence, in order to uphold the prestige of all concerned, it is again reiterated that the instructions be followed meticulously and judiciously.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>16</b> | The Examiners should acquaint themselves with the guidelines given in the “ <b>Guidelines for spot Evaluation</b> ” before starting the actual evaluation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>17</b> | Every Examiner shall also ensure that all the answers are evaluated, marks carried over to the title page, correctly totaled and written in figures and words.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>18</b> | The candidates are entitled to obtain photocopy of the Answer Book on request on payment of the prescribed processing fee. All Examiners/Additional Head Examiners/Head Examiners are once again reminded that they must ensure that evaluation is carried out strictly as per value points for each answer as given in the Marking Scheme.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

| MARKING SCHEME: PHYSICS(042) |                                                                                                                                 |       |             |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------|-------------|
| Code: 55/3/3                 |                                                                                                                                 |       |             |
| Q.No.                        | VALUE POINTS/EXPECTED ANSWERS                                                                                                   | Marks | Total Marks |
|                              | SECTION A                                                                                                                       |       |             |
| 1.                           | (a) Current in the circuit.                                                                                                     | 1     | 1           |
| 2.                           | (c) $\frac{h}{\lambda}$                                                                                                         | 1     | 1           |
| 3.                           | (c) $\frac{\mu_0 I^2}{2\pi r}$ , attractive                                                                                     | 1     | 1           |
| 4.                           | (d) 8:1                                                                                                                         | 1     | 1           |
| 5.                           | (a) Only displacement current exists.                                                                                           | 1     | 1           |
| 6.                           | (c) $I \propto A^2$                                                                                                             | 1     | 1           |
| 7.                           | (c) Width of depletion layer increases.                                                                                         | 1     | 1           |
| 8.                           | (c) <div style="text-align: center;">  </div> | 1     | 1           |
| 9.                           | (b) 0.51eV                                                                                                                      | 1     | 1           |
| 10.                          | (a) $\frac{V_0}{\sqrt{2}}$                                                                                                      | 1     | 1           |
| 11.                          | (c) $5 \times 10^9 \text{ m}^{-3}$                                                                                              | 1     | 1           |
| 12.                          | (b) twice                                                                                                                       | 1     | 1           |
| 13.                          | (b) $\frac{9}{5}$                                                                                                               | 1     | 1           |
| 14.                          | (c) $\frac{E}{3}$                                                                                                               | 1     | 1           |
| 15.                          | (b) inductor decreases and the capacitor increases.                                                                             | 1     | 1           |
| 16.                          | (c) Assertion (A) is true but Reason (R) is false                                                                               | 1     | 1           |
| 17.                          | (d) Assertion (A) is false and Reason (R) is also false                                                                         | 1     | 1           |
| 18.                          | (a) Both Assertion (A) and Reason(R) are true and Reason (R) is the correct explanation of the Assertion (A).                   | 1     | 1           |
|                              |                                                                                                                                 |       |             |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                           |   |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|---|
|     | <b>SECTION B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                           |   |
| 19. | <div> <div>Explanation of property</div> <div>Difference between half wave and full wave rectifier.</div> </div> <p>pn-junction conducts in forward bias.</p> <p><b>Alternatively:-</b>pn-junction is a uni-directional device.</p> <p>The half-wave rectifier gives output only for half of the input cycle .The full-wave rectifier gives output for both the halves of the input cycles.</p> <p><b>Alternatively:-</b>If output waveform of both the rectifiers shown diagrammatically, then full credit to be given.</p>                                                                                                                                                                                                                                                                                                                                                                                        | <div>1</div> <div>1</div> | 2 |
| 20. | <p>(a)</p> <div> <div>Graph of binding energy per nucleon as a function of mass number A</div> <div>Explanation</div> </div>  <p><b>Explanation:-</b>Nuclear forces are short range &amp; show saturation, while the electrostatic force are neither short range nor show any saturation. Hence for heavier nuclei (<math>A &gt; 170</math>) the electrostatic force of repulsion becomes predominant, decreasing the binding energy per nucleon.</p> <p><b>Alternatively:-</b>As the size of the nucleus increases, the nucleus becomes unstable.</p> <p><b>Note:</b> No deduction of marks if values of elements are not shown in the graph.</p> <p style="text-align: center;"><b>OR</b></p> <p>(b)</p> <div> <div>Expression for radius of the <math>n^{\text{th}}</math> orbit in a hydrogen atom</div> <div>2</div> </div> | <div>1</div> <div>1</div> |   |



|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                 |                            |             |      |                                                                          |   |  |  |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------|-------------|------|--------------------------------------------------------------------------|---|--|--|
|                                 | <p>Area of equilateral triangle <math>= \frac{\sqrt{3}}{4} a^2</math></p> <p><math>= \frac{\sqrt{3}}{4} \left( \frac{L^2}{9} \right) = \frac{\sqrt{3} L^2}{36}</math></p> <p>Magnetic moment <math>\propto</math> Area of the loop.</p> <p><math>\Rightarrow \frac{m_s}{m_i} = \frac{L^2}{16} \times \frac{36}{\sqrt{3} L^2} = \frac{3\sqrt{3}}{4}</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> | 2                          |             |      |                                                                          |   |  |  |
| 23.                             | <p>(a)</p> <table border="1"> <tr> <td>Statement of Huygen's Principle</td> <td>1</td> </tr> <tr> <td>Explanation</td> <td>1</td> </tr> </table> <p>Statement: Each point of the wavefront is the source of secondary disturbance in all directions.</p> <p>Common tangent to all the secondary wavelets gives new position of the wavefront.</p> <p>Explanation: Light energy cannot travel in backward direction.</p> <p><b>Alternatively:</b><br/>It was an adhoc assumption .</p> <p><b>Alternatively:</b><br/>For back wave: <math>I = \frac{1}{2} (1 + \cos \theta)</math><br/>at <math>\theta = 180^\circ</math> ; contribution is zero.</p> <p><b>Alternatively:</b><br/>Amplitude of secondary wavelets is maximum in forward direction and zero in backward in direction.</p> <p><b>Note:</b> If any other relevant explanation given, give full credit.</p> | Statement of Huygen's Principle                                                                 | 1                          | Explanation | 1    | <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p>1</p> |   |  |  |
| Statement of Huygen's Principle | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                 |                            |             |      |                                                                          |   |  |  |
| Explanation                     | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                 |                            |             |      |                                                                          |   |  |  |
|                                 | <p style="text-align: center;"><b>OR</b></p> <p>(b)</p> <table border="1"> <tr> <td>(i)</td> <td>Diagram for concave mirror</td> <td>1</td> </tr> <tr> <td>(ii)</td> <td>Diagram for convex lens</td> <td>1</td> </tr> </table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | (i)                                                                                             | Diagram for concave mirror | 1           | (ii) | Diagram for convex lens                                                  | 1 |  |  |
| (i)                             | Diagram for concave mirror                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1                                                                                               |                            |             |      |                                                                          |   |  |  |
| (ii)                            | Diagram for convex lens                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1                                                                                               |                            |             |      |                                                                          |   |  |  |

|     |                                                                                                                                                                                                                                                                                                      |                                                                  |   |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|---|
|     | <p>(i)</p>  <p>(ii)</p>                                                                                                          | 1                                                                |   |
|     |                                                                                                                                                                                                                                                                                                      | 1                                                                | 2 |
| 24. | <div> <div>Calculation of critical angle</div> <div>2</div> </div> <p><b>From Snell 's law:-</b><br/> <math>\mu_A \sin i_c = \mu_B \sin 90^\circ</math><br/> <math>2 \times \sin i_c = \sqrt{2} \times 1</math><br/> <math>\sin i_c = \frac{1}{\sqrt{2}}</math><br/> <math>i_c = 45^\circ</math></p> | $\frac{1}{2}$<br>$\frac{1}{2}$<br>$\frac{1}{2}$<br>$\frac{1}{2}$ |   |
|     | <p><b>Alternatively:</b><br/> <math>\sin i_c = \frac{1}{\mu_A}</math><br/> <math>\sin i_c = \frac{1}{\sqrt{2}}</math><br/> <math>i_c = 45^\circ</math></p>                                                                                                                                           | 1<br>$\frac{1}{2}$<br>$\frac{1}{2}$                              | 2 |

25.

Circuit Diagram

(a) Forward Biasing

$\frac{1}{2}$

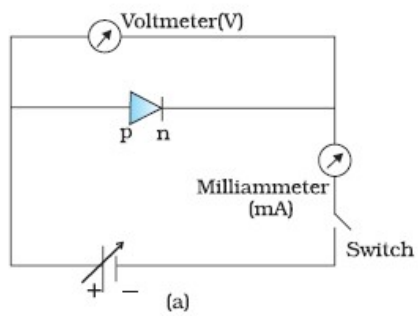
(b) Reverse Biasing

$\frac{1}{2}$

VI characteristics for each case

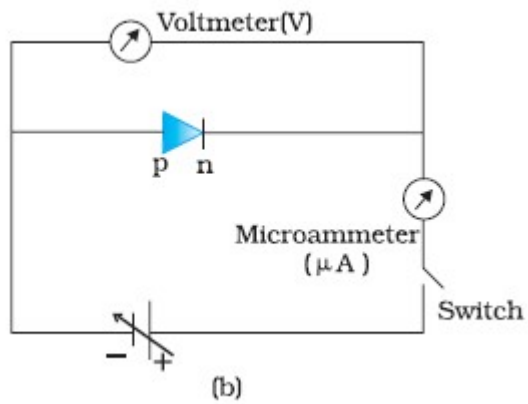
$\frac{1}{2} + \frac{1}{2}$

(a) Forward Bias



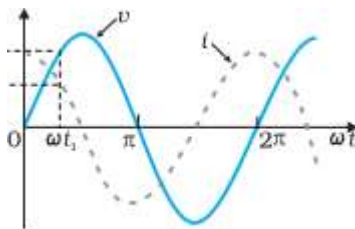
$\frac{1}{2}$

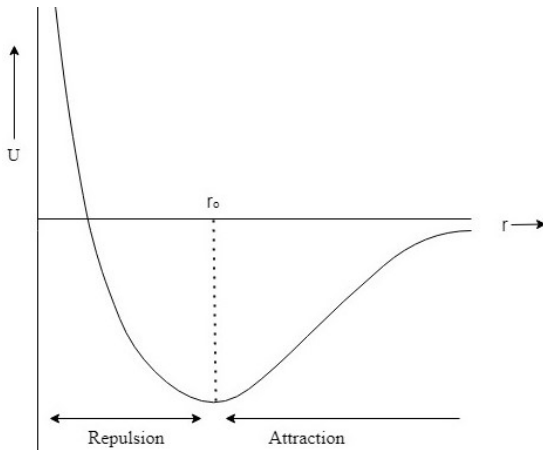
(b) Reverse Bias

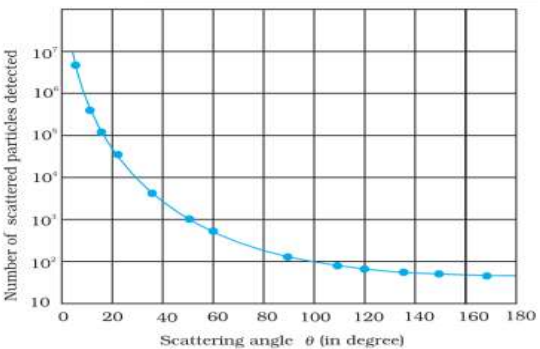


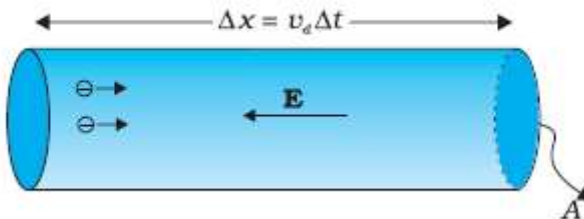
$\frac{1}{2}$

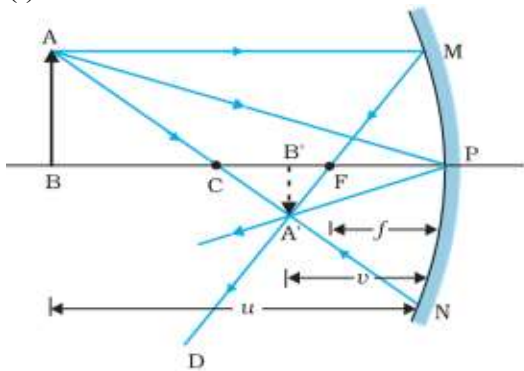


|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                       |   |                  |  |                              |     |                               |     |  |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|---|------------------|--|------------------------------|-----|-------------------------------|-----|--|--|
| <div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1                                     |   |                  |  |                              |     |                               |     |  |  |
| <div>OR</div> <div>(b)</div> <table border="1"><tr><td>Expression for average power consumed</td><td>2</td></tr><tr><td>Power Factor for</td><td></td></tr><tr><td>(i) Purely Inductive circuit</td><td>1/2</td></tr><tr><td>(ii) Purely Resistive Circuit</td><td>1/2</td></tr></table> <td></td> <td></td>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Expression for average power consumed | 2 | Power Factor for |  | (i) Purely Inductive circuit | 1/2 | (ii) Purely Resistive Circuit | 1/2 |  |  |
| Expression for average power consumed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2                                     |   |                  |  |                              |     |                               |     |  |  |
| Power Factor for                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                       |   |                  |  |                              |     |                               |     |  |  |
| (i) Purely Inductive circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1/2                                   |   |                  |  |                              |     |                               |     |  |  |
| (ii) Purely Resistive Circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1/2                                   |   |                  |  |                              |     |                               |     |  |  |
| <div>Instantaneous Power;</div> <div><math display="block">P = VI = (V_m \sin \omega t) \times i_m \sin(\omega t + \phi)</math></div> <div><math display="block">P = \frac{V_m i_m}{2} [\cos \phi - \cos(2\omega t + \phi)] \quad \text{----(1)}</math></div> <div>The average power over a cycle is given by the average of the two terms in the R.H.S of equation (1). It is only the second term which is time dependent .Its average is zero (the positive half of the cosine cancels the negative half).</div> <div>Therefore,</div> <div><math display="block">P_{avg} = \frac{V_m I_m}{2} \cos \phi = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos \phi</math></div> <div><math display="block">P_{avg} = V_{rms} I_{rms} \cos \phi</math></div> <div>Alternatively:-</div> <div>If the expression is deduced using integration, then full credit to be given.</div> | 1/2                                   |   |                  |  |                              |     |                               |     |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1/2                                   |   |                  |  |                              |     |                               |     |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1/2                                   |   |                  |  |                              |     |                               |     |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1/2                                   |   |                  |  |                              |     |                               |     |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1/2                                   |   |                  |  |                              |     |                               |     |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1/2                                   |   |                  |  |                              |     |                               |     |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1/2                                   |   |                  |  |                              |     |                               |     |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                       | 3 |                  |  |                              |     |                               |     |  |  |

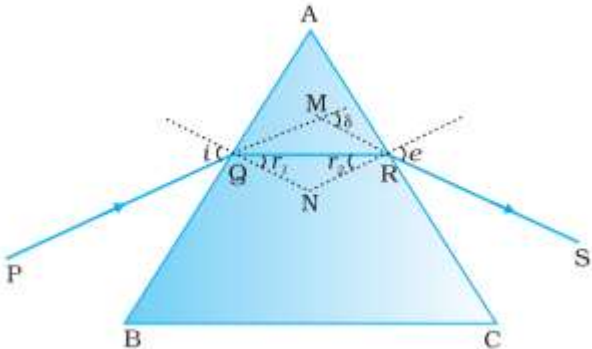
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |                                                                                                                                                                                               |  |  |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|
| 27.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |                                                                                                                                                                                               |  |  |  |
| (a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |                                                                                                                                                                                               |  |  |  |
| <div>(i) Prove that nuclear density is constant1</div> <div>(ii) Graph between potential energy &amp; separation1</div> <div>Two Inferences1/2 + 1/2</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |                                                                                                                                                                                               |  |  |  |
| <div>(i) <math>\rho = \frac{mass}{volume}</math></div> <div><math>= \frac{mas \quad number \times mass \ of \ nucleon}{volume \ of \ nucleus}</math></div> <div><math display="block">\rho = \frac{A \times m}{\frac{4}{3}\pi(R_0 A^{1/3})^3} = \frac{3m}{4\pi R_o^3}</math></div> <div>Hence, density is independent of mass number.</div> <div>(ii)</div> <div></div> <div>Inferences</div> <div><div>-The force is attractive for distances larger than r_0.</div><div>- The force is repulsive for distance less than r_0.</div></div> <div>Alternatively:-</div> <div>Any other relevant inference drawn from the graph should be given full credit.</div> <div>OR</div> <div>(b)</div> <tr><td colspan="2"><div>(i) Graph to show the variation of the number of scattered particles as a function of scattering angle.1</div><div>(ii) Two conclusions1/2 + 1/2</div><div>Discovery of nucleus1</div></td><td></td><td></td></tr> |  | <div>(i) Graph to show the variation of the number of scattered particles as a function of scattering angle.1</div> <div>(ii) Two conclusions1/2 + 1/2</div> <div>Discovery of nucleus1</div> |  |  |  |
| <div>(i) Graph to show the variation of the number of scattered particles as a function of scattering angle.1</div> <div>(ii) Two conclusions1/2 + 1/2</div> <div>Discovery of nucleus1</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |                                                                                                                                                                                               |  |  |  |

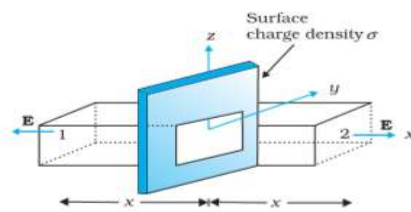
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                 |          |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|     | <p>(i)</p>  <p>(ii)- The entire positive charge and most of the mass of the atom are concentrated in a small space.<br/>       -Many of the <math>\alpha</math>-particles pass through the foil. It means that they do not suffer any collisions.</p> <p>To deflect the <math>\alpha</math>-particle backwards, a large repulsive force is required, which is provided only if the greater part of the mass of the atom &amp; its positive charge were concentrated tightly at its centre. This lead to the discovery of the nucleus in the atom.</p> | <p>1</p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p>1</p>                                                                                                               | <p>3</p> |
| 28. | <div style="border: 1px solid black; padding: 5px; margin-bottom: 10px;">           Finding the<br/>           (a) Current induced <span style="float: right;">1½</span><br/>           (b) Power required <span style="float: right;">1½</span> </div> <p>(a)</p> $i = \frac{Blv}{R}$ $= \frac{0.2 \times (10 \times 10^{-2}) \times (5 \times 10^{-2})}{0.4}$ $i = 2.5 \times 10^{-3} A$ <p>(b)</p> $P = i^2 R$ $= (2.5 \times 10^{-3})^2 \times 0.4$ $P = 2.5 \times 10^{-6} W$                                                                                                                                                      | <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> | <p>3</p> |

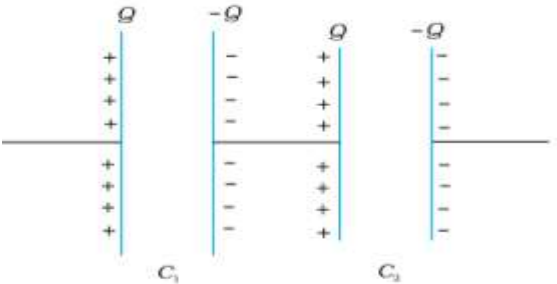
|                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                       |       |                        |    |                                     |   |                   |   |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|-------|------------------------|----|-------------------------------------|---|-------------------|---|
| 29.                                                   | <table><tr><td>Obtaining relation between current and drift velocity</td><td>1½</td></tr><tr><td>Deduction of Ohm's Law</td><td>1½</td></tr></table> <div></div> <p>Volume of the wire = <math>Al</math></p> <p>No. of <math>e^-</math> in this volume = <math>nAl</math></p> <p>Charge on the conductor = <math>(nAl)e</math></p> <p>The amount of charge crossing the area <math>A</math> in time <math>\Delta t</math> is <math>I\Delta t</math>; where <math>I</math> is the magnitude of the current. Hence,</p> $I\Delta t = +neA v_d \Delta t$ $\Rightarrow I = neAv_d$ <p>Since <math> v_d  = \frac{eE}{m}\tau</math></p> $I = neA\left(\frac{e\tau}{m}\right)\frac{V}{l} = \left(\frac{ne^2\tau}{m}\right)A\left(\frac{V}{l}\right)$ $\Rightarrow \frac{V}{I} = \left(\frac{m}{ne^2\tau}\right)\frac{l}{A}$ <p>Hence <math>\frac{V}{I}</math> is constant for a given conductor at constant temperature.</p> $\Rightarrow V \propto I$ | Obtaining relation between current and drift velocity | 1½    | Deduction of Ohm's Law | 1½ | <p>½</p> <p>½</p> <p>½</p> <p>½</p> |   |                   |   |
| Obtaining relation between current and drift velocity | 1½                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                       |       |                        |    |                                     |   |                   |   |
| Deduction of Ohm's Law                                | 1½                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                       |       |                        |    |                                     |   |                   |   |
| 30.                                                   | <table><tr><td>(a) (i) Identification of heavier mass &amp; reason</td><td>½ + ½</td></tr><tr><td>(ii) Slope of line</td><td>1</td></tr><tr><td>(b) Calculation of momentum</td><td>1</td></tr></table> <p>(a)(i) B is heavier than A</p> <p>Reason- Slope <math>\propto \frac{1}{\sqrt{m}}</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | (a) (i) Identification of heavier mass & reason       | ½ + ½ | (ii) Slope of line     | 1  | (b) Calculation of momentum         | 1 | <p>½</p> <p>½</p> | 3 |
| (a) (i) Identification of heavier mass & reason       | ½ + ½                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                       |       |                        |    |                                     |   |                   |   |
| (ii) Slope of line                                    | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                       |       |                        |    |                                     |   |                   |   |
| (b) Calculation of momentum                           | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                       |       |                        |    |                                     |   |                   |   |

|                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                 |   |                                          |   |                               |   |                        |   |   |  |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---|------------------------------------------|---|-------------------------------|---|------------------------|---|---|--|
|                                                                      | <p>Slope of A &gt; slope of B;</p> <p>Hence <math>m_B &gt; m_A</math></p> <p>(ii) <math>\lambda = \frac{h}{\sqrt{2mqV}}</math></p> $\lambda = \frac{h}{\sqrt{2mq}} \times \frac{1}{\sqrt{V}}$ <p>So, slope = <math>\frac{h}{\sqrt{2mq}}</math></p> <p>(b) Momentum, <math>p = \frac{h}{\lambda}</math></p> $= \frac{6.63 \times 10^{-34}}{3 \times 10^{-10}}$ $p = 2.21 \times 10^{-24} \text{ kgm/s}$                                                                                                                                                                              | <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> | 3 |                                          |   |                               |   |                        |   |   |  |
| SECTION D                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                 |   |                                          |   |                               |   |                        |   |   |  |
| 31.                                                                  | <p>(a)</p> <table><tr><td>(i) Ray diagram showing formation of real image in a concave mirror.</td><td>1</td></tr><tr><td>Obtaining the relation between u,v and R</td><td>2</td></tr><tr><td>(ii) Position of image formed</td><td>1</td></tr><tr><td>Height of image formed</td><td>1</td></tr></table> <p>(i)</p>  <p>From Fig. the two right-angled triangles A'B'F and MPF are similar. (For paraxial rays, MP can be considered to be a straight line perpendicular to CP.) Therefore,</p> | (i) Ray diagram showing formation of real image in a concave mirror.                                                            | 1 | Obtaining the relation between u,v and R | 2 | (ii) Position of image formed | 1 | Height of image formed | 1 | 1 |  |
| (i) Ray diagram showing formation of real image in a concave mirror. | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                 |   |                                          |   |                               |   |                        |   |   |  |
| Obtaining the relation between u,v and R                             | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                 |   |                                          |   |                               |   |                        |   |   |  |
| (ii) Position of image formed                                        | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                 |   |                                          |   |                               |   |                        |   |   |  |
| Height of image formed                                               | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                 |   |                                          |   |                               |   |                        |   |   |  |

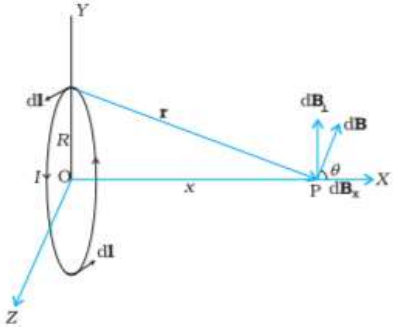
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                 |  |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | $\frac{B'A'}{PM} = \frac{B'F}{FP}$ $\text{or } \frac{B'A'}{BA} = \frac{B'F}{FP} \quad (\because PM = AB) \quad \text{-----(i)}$ <p>Since <math>\angle APB = \angle A'PB'</math>, the right angled triangles <math>A'B'P</math> and <math>ABP</math> are also similar. Therefore,</p> $\frac{B'A'}{BA} = \frac{B'P}{BP} \quad \text{-----(ii)}$ <p>Comparing equations (i) and (ii)</p> $\frac{B'F}{FP} = \frac{B'P - FP}{FP} = \frac{B'P}{BP} \quad \text{----- (iii)}$ <p><math>B'P = -v</math>, <math>FP = -f</math>, <math>BP = -u</math>;</p> <p>Using these in Eq.(iii) we get <math>\frac{1}{v} + \frac{1}{u} = \frac{1}{f} = \frac{2}{R}</math></p> <p><b>Alternatively:-</b> If the result derived by any other method, full credit to be given.</p> <p>(ii) For lens: <math>\frac{1}{v} - \frac{1}{u} = \frac{1}{f}</math></p> <p><math>u = -5m</math>; <math>f = +1m</math></p> $\frac{1}{v} - \frac{1}{-5} = \frac{1}{(+1)}$ $\Rightarrow v = \frac{5}{4}m = 1.25m$ $m = \frac{I}{O} = \frac{v}{u} = \frac{\left(+\frac{5}{4}\right)}{(-5)}$ <p><math>I = (-0.25) \times (1.8)</math><br/> <math>I = -0.45 \text{ m}</math></p> | <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> |  |
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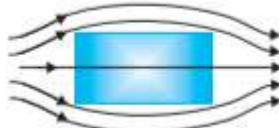
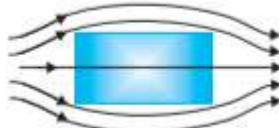
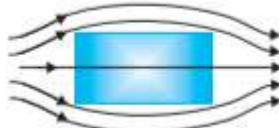
| OR                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| (b)                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |
| <div style="border: 1px solid black; padding: 5px;">           (i) Ray diagram showing refraction of a ray of light through a rectangular glass prism. <span style="float: right;">1</span><br/>           Obtaining the relation between <math>\mu, A</math> &amp; <math>\delta_m</math> <span style="float: right;">2</span><br/>           (ii) Finding Refractive Index of material of the lens. <span style="float: right;">2</span> </div> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |
| (i)                                                                                                                                                                                                                                                                                                                                                                                                                                              | <div style="text-align: center;">  </div> <p>In the quadrilateral AQNR, two of the angles (at the vertices Q and R) are right angles. Therefore, the sum of the other angles of the quadrilateral is <math>180^\circ</math>.<br/> <math>\angle A + \angle QNR = 180^\circ</math><br/> From the triangle QNR, <math>r_1 + r_2 + \angle QNR = 180^\circ</math><br/> Comparing these two equations, we get<br/> <math>r_1 + r_2 = A</math> -----(i)<br/> The total deviation <math>\delta</math> is the sum of deviations at the two faces,<br/> <math>\delta = (i - r_1) + (e - r_2)</math> that is, <math>\delta = i + e - A</math> -----(ii)</p> <p>When <math>\delta = \delta_m</math>; <math>i = e</math> &amp; <math>r_1 = r_2</math><br/> From (i); <math>2r = A</math> or <math>r = A/2</math></p> <p>From (ii); <math>\delta_m = 2i - A</math> or <math>i = \frac{A + \delta_m}{2}</math></p> $\mu = \frac{\sin i}{\sin r} = \frac{\sin(\frac{A + \delta_m}{2})}{\sin \frac{A}{2}}$ | 1             |
| (ii) Given; P = - 5D                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\frac{1}{2}$ |
| $f$ (in cm) = $\frac{100}{(-5)} = - 20$ cm                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\frac{1}{2}$ |
| Using Lens Maker's formula ; $\frac{1}{f} = (\mu - 1)[\frac{1}{R_1} - \frac{1}{R_2}]$                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\frac{1}{2}$ |
| $\frac{1}{(-20)} = (\mu - 1)[\frac{1}{(-20)} - \frac{1}{(+20)}]$                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |

|                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                           |                             |                                                                                                  |   |                                       |   |                                                                              |  |
|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-----------------------------|--------------------------------------------------------------------------------------------------|---|---------------------------------------|---|------------------------------------------------------------------------------|--|
|                                                                                                  | $\frac{1}{(-20)} = (\mu - 1) \left[ -\frac{1}{10} \right]; \quad \mu - 1 = \frac{1}{2}$ $\Rightarrow \mu = \frac{3}{2} = 1.5$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\frac{1}{2}$<br><br>$\frac{1}{2}$        | 5                           |                                                                                                  |   |                                       |   |                                                                              |  |
| 32.                                                                                              | <p>(a)</p> <table border="1"> <tr> <td>(i) Definition &amp; SI Unit of Electric Flux</td> <td><math>\frac{1}{2} + \frac{1}{2}</math></td> </tr> <tr> <td>(ii) Deriving the expression for electric field due to a uniformly charged infinite plane sheet.</td> <td>2</td> </tr> <tr> <td>(iii) Net charge enclosed by the cube</td> <td>2</td> </tr> </table> <p>(i) <math>\phi = \vec{E} \cdot \vec{A}</math><br/> <b>Alternatively:</b> Electric flux is the number of electric field lines passing through an area normally.<br/> S.I. unit of electric flux <math>\text{Nm}^2/\text{C}</math> or <math>\text{V-m}</math>.</p> <p>(ii)</p>  <p>From Gauss's law:- <math>\phi = \oint \vec{E} \cdot d\vec{A} = \frac{q}{\epsilon_0}</math></p> $2EA = \frac{\sigma A}{\epsilon_0}$ $E = \frac{\sigma}{2\epsilon_0}$ <p><b>Alternatively:</b> If the shape of the Gaussian surface is taken cylindrical, full credit to be given.</p> | (i) Definition & SI Unit of Electric Flux | $\frac{1}{2} + \frac{1}{2}$ | (ii) Deriving the expression for electric field due to a uniformly charged infinite plane sheet. | 2 | (iii) Net charge enclosed by the cube | 2 | $\frac{1}{2}$<br><br>$\frac{1}{2}$<br><br>$\frac{1}{2}$<br><br>$\frac{1}{2}$ |  |
| (i) Definition & SI Unit of Electric Flux                                                        | $\frac{1}{2} + \frac{1}{2}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                           |                             |                                                                                                  |   |                                       |   |                                                                              |  |
| (ii) Deriving the expression for electric field due to a uniformly charged infinite plane sheet. | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                           |                             |                                                                                                  |   |                                       |   |                                                                              |  |
| (iii) Net charge enclosed by the cube                                                            | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                           |                             |                                                                                                  |   |                                       |   |                                                                              |  |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                   |                             |                                                         |   |                                                     |   |                                                                                                                                                                                                                                                                 |  |
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| <p>(iii)</p> $\phi_L = Eds \cos 180^\circ = -Eds$ $= -BL^2$ $\phi_R = Eds \cos 0^\circ = Eds$ $= (AL + B)L^2 = AL^3 + BL^2$ <p>Net flux <math>= \phi_L + \phi_R</math></p> $= (AL^3 + BL^2) - BL^2$ <p>Net flux <math>= AL^3 = \frac{q}{\epsilon_o}</math></p> $\Rightarrow q = AL^3 \epsilon_o$ <p style="text-align: center;"><b>OR</b></p> <p>(b)</p> <table border="1"><tr><td>(i) Definitions &amp; S.I. Unit of electric potential</td><td><math>\frac{1}{2} + \frac{1}{2}</math></td></tr><tr><td>(ii) Derivation of expression of Equivalent capacitance</td><td>2</td></tr><tr><td>(iii) Calculation of Electrostatic Potential Energy</td><td>2</td></tr></table> <p>(i) Electrical Potential – Electrostatic potential at any point in a region with electrostatic field is the work done in bringing a unit positive charge (without acceleration) from infinity to that point.</p> <p><b>Alternatively:-</b></p> $V = \frac{\text{Work Done}}{q}$ $V = -\int \vec{E} \cdot d\vec{l}$ <p>S.I. unit of electrostatic potential is volt.</p> <p><b>Alternatively:-</b><br/>S.I. unit is J/C.</p> <div></div> | (i) Definitions & S.I. Unit of electric potential | $\frac{1}{2} + \frac{1}{2}$ | (ii) Derivation of expression of Equivalent capacitance | 2 | (iii) Calculation of Electrostatic Potential Energy | 2 | <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> |  |
| (i) Definitions & S.I. Unit of electric potential                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $\frac{1}{2} + \frac{1}{2}$                       |                             |                                                         |   |                                                     |   |                                                                                                                                                                                                                                                                 |  |
| (ii) Derivation of expression of Equivalent capacitance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2                                                 |                             |                                                         |   |                                                     |   |                                                                                                                                                                                                                                                                 |  |
| (iii) Calculation of Electrostatic Potential Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2                                                 |                             |                                                         |   |                                                     |   |                                                                                                                                                                                                                                                                 |  |



|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                    |  |
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|  | <p>circuit.<br/>-It is a sensitive device.<br/>(Any one of the above)</p> <p>(ii) The magnetic field is made radial in a moving coil galvanometer so that the magnetic dipole moment (<math>\vec{m}</math>) is always perpendicular to the magnetic field (<math>\vec{B}</math>) Hence, <math>\sin \theta = 1</math> always<br/><b>Alternatively:</b> The magnetic field is made radial in a moving coil galvanometer to make the scale linear.</p> <p>It is achieved by using curved magnetic poles.</p> <p><b>Alternatively:-</b>By using soft iron cylindrical core.</p> <p style="text-align: center;"><b>OR</b></p> <p>(b)</p> <div style="border: 1px solid black; padding: 5px; margin: 10px 0;"> <p>(i) Derivation of expression for magnetic field on the axis of a current carrying loop. <span style="float: right;">3</span></p> <p>(ii) Two differences between diamagnetic and paramagnetic substance. <span style="float: right;">1+1</span></p> </div> <p>(i)</p>  $dB = \frac{\mu_0}{4\pi} \frac{I  d\vec{l} \times \vec{r} }{r^3}$ $= \frac{\mu_0 i dl \sin 90^\circ}{4\pi (x^2 + R^2)^{3/2}}$ <p><math>dB_{\perp}</math> cancels out.</p> <p>Net <math>B = \int dB_x = \int dB \cos \theta</math></p> | <p>1</p> <p>1</p> <p>1</p> <p></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> |  |
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| $= \frac{\mu_o}{4\pi} \int \frac{idl}{(x^2 + R^2)} \times \frac{R}{(x^2 + R^2)^{1/2}}$ $= \frac{\mu_o i R}{4\pi (x^2 + R^2)^{3/2}} \int dl$ $= \frac{\mu_o i R}{4\pi (x^2 + R^2)^{3/2}} (2\pi R)$ $\mathbf{B} = B_x \hat{\mathbf{i}} = \frac{\mu_o I R^2}{2(x^2 + R^2)^{3/2}} \hat{\mathbf{i}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $\frac{1}{2}$<br>$\frac{1}{2}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |
| <p>(ii) Differences</p> <table><tr><td><p><b>Diamagnetic Materials</b></p><p>(i) Susceptibility is between -1 and 0.</p><p>(ii) Relative permeability is between 0 and 1.</p><p>(iii) <math>\mu &lt; \mu_o</math></p><p>(iv) Tendency to move from stronger to weaker part of external magnetism.</p><p>(v) is repelled by a magnet.</p><p>(vi) Field inside the material is reduced.</p><p>(vii)</p></td><td><p><b>Paramagnetic Materials</b></p><p>(i) Susceptibility is a small positive number.(slightly greater than zero.)</p><p>(ii) Relative permeability is slightly greater than 1.</p><p>(iii) <math>\mu &gt; \mu_o</math></p><p>(iv) Tendency to move from region of weak to strong magnetic field.</p><p>(v) is weakly attracted by a magnet.</p><p>(vi) Field inside is slightly enhanced.</p><p>(vii)</p></td><td>1+1</td></tr></table> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     | <p><b>Diamagnetic Materials</b></p> <p>(i) Susceptibility is between -1 and 0.</p> <p>(ii) Relative permeability is between 0 and 1.</p> <p>(iii) <math>\mu &lt; \mu_o</math></p> <p>(iv) Tendency to move from stronger to weaker part of external magnetism.</p> <p>(v) is repelled by a magnet.</p> <p>(vi) Field inside the material is reduced.</p> <p>(vii)</p>  | <p><b>Paramagnetic Materials</b></p> <p>(i) Susceptibility is a small positive number.(slightly greater than zero.)</p> <p>(ii) Relative permeability is slightly greater than 1.</p> <p>(iii) <math>\mu &gt; \mu_o</math></p> <p>(iv) Tendency to move from region of weak to strong magnetic field.</p> <p>(v) is weakly attracted by a magnet.</p> <p>(vi) Field inside is slightly enhanced.</p> <p>(vii)</p>  | 1+1 |
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| <p>Any two of the above mentioned differences.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |
| <p>5</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |

|     | SECTION E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                 |  |
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| 34. | <div style="border: 1px solid black; padding: 5px; margin-bottom: 10px;">           (a) Time taken by the electron to strike the edge. 1<br/>           (b) Shape of path followed by the electron and it's reason <math>\frac{1}{2} + \frac{1}{2}</math><br/>           (c) Potential Difference applied 2<br/>           OR<br/>           (c) Magnitude and Direction of magnetic field 1+1         </div> <p>(a) Electron strikes the edge after travelling 3 cm horizontally (along x-axis).<br/> <math>S_x = v_x \times t</math><br/> <math>3 \times 10^{-2} = (3 \times 10^7) \times t</math><br/> <math>t = 10^{-9} \text{ s}</math></p> <p>(b) Shape of the path is parabola.<br/>           Reason: Force/acceleration is in a fixed direction perpendicular to the initial velocity.</p> <p>(c) Along y-direction<br/> <math>S_y = u_y t + \frac{1}{2} a_y t^2</math><br/> <math>-0.5 \times 10^{-2} = 0 + \frac{1}{2} a_y (10^{-9})^2</math><br/> <math>a_y = -10^{16} \text{ m/s}^2</math><br/>           Magnitude of acceleration <math>(a_y) = \frac{eE}{m} = \frac{e}{m} \left( \frac{V}{l} \right)</math><br/> <math>V = \frac{10^{16} \times 9.1 \times 10^{-31} \times 1 \times 10^{-2}}{1.6 \times 10^{-19}}</math><br/> <math>V = 568.75 \text{ V}</math></p> <p style="text-align: center;"><b>OR</b></p> <p>(c) <math> qE  =  qvB </math>; <math>B = \frac{E}{v} = \left( \frac{ma_y}{e} \right) \left( \frac{1}{v} \right)</math><br/>           Along y-direction<br/> <math>S_y = u_y t + \frac{1}{2} a_y t^2</math><br/> <math>-0.5 \times 10^{-2} = 0 + \frac{1}{2} a_y (10^{-9})^2</math><br/> <math>a_y = -10^{16} \text{ m/s}^2</math><br/> <math>B = \left( \frac{9.1 \times 10^{-31} \times 10^{16}}{1.6 \times 10^{-19}} \right) \times \left( \frac{1}{3 \times 10^7} \right)</math></p> | <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> |  |



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|  | <p>Any two of the above differences.</p> <p style="text-align: center;"><b>OR</b></p> <p>(c) The opening (slit) is 3m; which is of the order of the wavelength of sound waves whereas it is very large compare to the wavelength of light.<br/>Hence, sound can bend around the obstacle while light cannot.</p> | 2 | 4 |
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